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Exploring the feature prioritization and data sampling of PCOS diagnosis via densely connected attention based squeeze deep learning detection model

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ABSTRACT

Accurate and timely diagnosis remains a challenge due to the complexity of Polycystic Ovary Syndrome (PCOS) symptoms and data imbalance issues in the existing datasets. This research aims to develop a robust PCOS detection model that addresses these challenges by introducing a novel hybrid methodology with effective feature prioritization while handling data balancing issues. The research involves three major phases: pre-processing, feature selection, and PCOS detection. In the pre-processing stage, dataset balancing is emphasized by the combination of Synthetic Minority Oversampling Techniques (SMOTE) and Edited Nearest Neighbor (ENN). Under this stage, replacing null values, balancing the dataset, and dropping unnecessary columns are accomplished to increase PCOS detection accuracy. The second stage is feature selection, where a distinct hybrid bionic strategy named the Gorilla Salp Swarm Troop Model (GS2TM) is proposed to pick the optimal set of dominant features. The GS2TM algorithm reduces the feature set by 51.1 %, retaining only 23 features while achieving a state-of-the-art accuracy of 98.7 %. In addition, the Densely Connected Attention-Based Squeeze Convolutional Detection Model (DASCD) is proposed for the prediction of PCOS, in which multiple layers are adjusted in a feed-forward manner. The novelty of this work lies in the unified pipeline that simultaneously addresses three major challenges in PCOS detection, such as dataset imbalance (SMOTE-ENN), feature redundancy (GS2TM), and overfitting (DASCD with attention), providing both high accuracy and enhanced interpretability. As a result, the proposed detection model greatly improves accuracy compared to other existing ML-based strategies. Specifically, by utilizing 23 characteristics with GS2TM, the proposed model outperforms with an accuracy of 98.7 % in categorizing PCOS and non-PCOS.

1. Introduction

Polycystic Ovary Syndrome (PCOS) is a prevalent endocrine disorder that affects 5–10 % of women of reproductive age (15–45 years), often leading to infertility, metabolic complications, and long-term risks such as cardiovascular disease and type 2 diabetes [1–9]. Conventional diagnostic approaches, including ultrasound imaging, hormone profiling, and clinical assessments, though widely used, are resource-intensive and subject to inter-observer variability. Recent research has explored machine learning (ML) and deep learning (DL) as alternatives for PCOS detection, showing promise in handling large datasets, discovering hidden patterns, and improving predictive accuracy [10–15,30,35–37]. Previous studies have demonstrated that ML and DL can aid in PCOS detection by integrating clinical and imaging data, applying feature selection, and leveraging classifiers such as SVM, Random Forests, and deep CNNs. These models highlight the potential of artificial intelligence in improving diagnostic accuracy and reducing manual dependency in clinical workflows.

Despite these advancements, significant limitations persist. Most existing models are trained on imbalanced and heterogeneous datasets, leading to bias and poor generalization. High-dimensional feature spaces often introduce redundancy, increasing the risk of overfitting. Many models function as "black boxes," offering little interpretability, which restricts clinical trust and adoption. Moreover, external validations using multi-center or prospective datasets are scarce, making clinical applicability uncertain. Despite these advancements, significant limitations persist. Most existing models are trained on imbalanced and heterogeneous datasets, leading to bias and poor generalization. High-

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dimensional feature spaces often introduce redundancy, increasing the risk of overfitting. Many models function as "black boxes," offering little interpretability, which restricts clinical trust and adoption. Moreover, external validations using multi-center or prospective datasets are scarce, making clinical applicability uncertain. Recent developments in related medical-imaging domains highlight complementary directions. For example, ensembles optimized with evolutionary strategies and swarm-based optimizers have improved robustness in medical-image classification tasks [38], while lightweight attention-based CNNs [39] and federated/lightweight designs [40] have advanced deployment and privacy-preserving model sharing in clinical settings. These advances motivate a combined focus on (i) advanced feature selection to reduce redundancy while preserving discriminative features, (ii) robust data balancing to mitigate bias, (iii) interpretable models that provide clinical insights, and (iv) validation frameworks to support generalization across diverse populations.

This study addresses these needs through a unified pipeline that integrates SMOTE-ENN for data balancing, GS2TM for optimized feature selection, and DASCD for deep attention-based classification, coupled with SHAP analysis to enhance interpretability. The proposed GS2TM introduces a hybrid metaheuristic strategy that integrates the exploration strength of the Gorilla Troop Optimizer with the exploitation efficiency of the Salp Swarm Algorithm. Hybrid and swarm-based optimizers have recently shown strong performance for optimization and feature-selection tasks in medical applications, motivating our two-level cooperative design [41,42]. Unlike conventional metaheuristics, GS2TM employs a two-level cooperative mechanism where gorilla-based exploration dynamically guides salp swarm exploitation, while a troop memory updating strategy mitigates premature convergence and improves search diversity. This design enables effective dimensionality reduction while preserving the most discriminative features, which is not attainable by existing single-heuristic methods.

The DASCD further strengthens the novelty of this work. It extends beyond generic CNN-attention frameworks by combining a densely connected convolutional backbone with attention-driven squeeze operations, specifically tailored to capture subtle interdependencies among PCOS-related clinical and ultrasound features. Recent work on lightweight attention mechanisms [39] and federated-friendly architectures [40] demonstrates the utility of compact, attention-aware designs for robust clinical deployment and multi-site collaboration, which informed our architecture choices. The adaptive feature recalibration module highlights medically relevant attributes while suppressing redundant ones, thereby improving interpretability and clinical trustworthiness. When integrated with SMOTE-ENN for dataset balancing, the unified pipeline using SMOTE-ENN, GS2TM, and DASCD provides a comprehensive solution to three major challenges in PCOS detection: dataset imbalance, feature redundancy, and overfitting, establishing a unique and clinically meaningful framework. Recent developments have emphasized the significance of explainability and generalization across multi-site datasets in order to guarantee that the pipeline is clinically useful [43]. By pairing DASCD with SHAP, our framework bridges computational accuracy with interpretability, supporting adoption in clinical workflows.

The main contributions of this study are as follows:

- ➢ A hybrid data balancing approach is introduced by integrating SMOTE with ENN to address class imbalance in PCOS datasets, improving generalization and reducing bias.
- ➢ A new GS2TM is proposed, which combines the exploration strength of Gorilla Troop Optimizer with the exploitation efficiency of the Salp Swarm Algorithm, enabling effective dimensionality reduction and retention of discriminative features.
- ➢ A novel DASCD model is developed to capture spatial, temporal, and channel-wise dependencies in clinical and ultrasound features. Its dense connections support feature reuse, squeeze layers provide compact representations, and attention mechanisms highlight medically relevant attributes to enhance interpretability.
- ➢ SHAP is applied to ensure transparency, offering insights into the contribution of each feature toward predictions and bridging the gap between computational performance and clinical trust.
- ➢ The unified pipeline that integrates SMOTE-ENN, GS2TM, and DASCD collectively addresses key challenges of dataset imbalance, feature redundancy, and overfitting, while enhancing interpretability and clinical relevance.

1.1. Background of the study

PCOS affects 5–10 % of women of reproductive age and is marked by irregular menstruation, hormonal imbalance, and multiple ovarian cysts. These manifestations often cause infertility, metabolic complications, and elevated risks of cardiovascular disease and type 2 diabetes. Standard diagnosis relies on ultrasound imaging, hormone assays, and clinical evaluation, which can be resource-intensive and time-consuming. Recent advances in ML and DL have shown potential for automating PCOS detection, yet imbalanced datasets, inadequate preprocessing, and limited interpretability constrain most existing studies. The absence of broad clinical validation and the underutilization of optimization methods further restricts their real-world applicability. The motivation for this study lies in addressing these gaps by developing an automated and interpretable PCOS detection system that not only enhances predictive accuracy but also improves clinical trust and usability. To achieve this, the proposed framework integrates both clinically intensive features and easily accessible indicators. This design aims to reduce diagnostic delays, support early risk stratification, and extend PCOS detection capabilities to primary care and resource-limited settings where comprehensive endocrinological testing is not always feasible.

The research paper is organized as follows: Section 2 describes the recent study related to PCOS detection with its problem statement. Section 3 describes the proposed work, including several stages and network architectures. Sections 4 and 5 describe the experimental setup, results and discussion. Sections 6 and 7 end with the conclusion and future scope.

2. Literature survey

Some of the existing literature related to the proposed methodology is reviewed here.

PCOS health disorder affects around 10 million women worldwide, and in India, one in five women is affected by it due to a lack of awareness. If PCOS is not diagnosed and managed at an early stage, it can lead to various complications, including type 2 diabetes, cardiovascular disease, hypertension, infertility, and other metabolic disorders. So, it is crucial to detect the disease at an early stage. Aggarwal and Pandey [16] performed the data amalgamation technique, which was performed on disease recognized datasets. Supervised and unsupervised machine learning methods were applied to evaluate the hidden relationship of PCOS with other disorders. Significant features of other diseases were obtained to recognize the PCOS disorder, and it gave the best results with significant features.

PCOS is an endocrinological abnormality and leads to the formation of ovarian cysts, persistent disorders of hormonal secretion, and serious health complications. It is very crucial to identify PCOS in real-world clinical detection techniques. For real-time diagnosis, the authors required an AI-based PCOS prediction model, which is considered an error-prone technique. Sayma and Muhammad [17] developed a modified ensemble ML classification method by considering existing stacking methods for PCOS recognition with patient symptom data. Five traditional machine learning methods were used for the base learning process, and a boosting or bagging ensemble learning model was used as the meta-learner of the stacked model. This stacking ensemble technique achieved better classification accuracy, combined with the existing methods.

High levels of male hormones and a delayed menstrual cycle are the common symptoms of PCOS disorder, affecting a huge percentage of women in the reproductive stage. This also creates a number of small fluid-filled sacs known as follicles in both ovaries of women to facilitate the frequent egg discharge. The risk of long-term complications is reduced by early PCOS disease. Shamik Tiwari et al. [18] focused on diagnosing PCOS based on the clinical dataset obtained from the Kaggle repository. The Random Forest (RF) approach obtained an accuracy of 93.25 % which is higher than other existing approaches.

Varada Vivek Khanna et al. [19] suggested different DL and ML methods to predict PCOS between fertile patients. This research utilized the publicly available dataset of 541 patients from Kerala, India. This study's main objective was to identify PCOS in patients while creating an automated method of screening that used a machine learning algorithm to assist doctors in making decisions. The trustworthy, interpretable, and understandable prediction was obtained by the explainable AI (XAI) technique.

Most women of reproductive age are affected by a common disease, PCOS, which leads to metabolic and hormonal disorders. Baweja and Kanchana [20] developed an accurate and cost-effective predictor that obtained only the accurate features for further recognition processes. Different ML methods like Multi-layer Perceptron (MLP), Support Vector Machine (SVM), Logistic regression, Naïve Bayes, and Classification and Regression Tree (CART) methods were utilized to perform the prediction process. From the result analysis, the MLP achieved better classification accuracy and exponentially minimized the computational time and cost associated with the prediction process.

A hybrid framework that combined deep learning (DL) and classic machine learning (ML) methods was suggested by Mridul et al. [28] to predict PCOS based on biochemical and clinical parameters. Support Vector Machines (SVM), Decision Trees, Random Forests, and Logistic Regression were the machine learning (ML) approaches used in the study for feature selection and subsequent classification. The accuracy of prediction was increased by using deep learning models, such as Artificial Neural Networks (ANNs), to capture non-linear interactions. Their results showed that, when dealing with complicated and high-dimensional data, DL models outperformed conventional ML techniques. However, pre-processing methods like data imputation and normalization were used to overcome issues like missing values and dataset imbalance.

Panda et al. [25] focused on integrating clinical and biochemical data to improve diagnostic precision while developing and evaluating machine learning (ML) models for the identification of PCOS. In order to categorize PCOS and pinpoint important diagnostic characteristics, the study assessed the effectiveness of many machine learning techniques, such as Decision Trees, Support Vector Machines (SVM), Random Forests, and Gradient Boosting. The authors demonstrated the value of feature selection techniques like Recursive Feature Elimination (RFE) in enhancing model efficiency through thorough model analysis. By using oversampling techniques like SMOTE (Synthetic Minority Oversampling Technique), the study also addressed typical issues like data imbalance and placed an emphasis on pre-processing strategies to deal with noisy and missing data.

A web-based platform that incorporated machine learning (ML) techniques was presented by Rahman et al. [26] to aid in the early diagnosis and prediction of PCOS. Clinical, lifestyle, and biochemical data were analyzed using a combination of machine learning methods, such as Random Forest, Gradient Boosting, and Logistic Regression. Data pre-processing methods, including feature normalization and missing value imputation, were used to improve model performance. The platform revealed important risk variables linked to PCOS, including irregular menstrual cycles and hormonal abnormalities. It showed good prediction accuracy, especially when using ensemble techniques like Gradient Boosting. In order to fill deficiencies in conventional healthcare delivery systems, the author highlighted the possibility of combining machine learning (ML) with online technologies to offer scalable and easily accessible options for early PCOS detection.

Earlier detection of PCOS preserves fertility in women, avoiding long term health complications, including obesity, heart disease, and hypertension. The objective of Faris et al. [31] is to extract the most effective PCOS features that support experts in early diagnosis. For that, the feature extraction model KM-GN has been introduced. It integrated the k-means technique with a genetic approach for detecting the most informative features for PCOS detection. The selected features are fed to Random Subspace-based Bootstrap Aggregating Ensembles (RSBE), and these results are compared with individual and ensemble classifiers. The statistical metrics, including the confusion matrix, accuracy, and precision, were evaluated, and better performance was noted.

Emara et al. [33] developed ML assisted framework for classifying PCOS instances. This approach includes stacked learning and is based on the Adaptive Synthetic (ADASYN) approach, random oversampling, and Synthetic Minority Over-sampling Technique (SMOTE) for the data imbalance issue. The BORUTA approach was utilized for feature selection with the aim of improving precision and other classification performance metrics. This feature selection provided 97 % accuracy for PCOS classification.

However, despite these advancements, several limitations persist. Most existing studies rely on small or region-specific datasets, limiting generalizability. Many approaches either do not explicitly handle feature prioritization or suffer from high dimensionality and redundancy, leading to potential overfitting. Few methods provide an integrated solution that combines robust feature selection with a highly expressive prediction model, while maintaining interpretability and clinical relevance. To overcome these gaps, the proposed work introduces a unified pipeline that integrates SMOTE-ENN for dataset balancing, the GS2TM for dominant feature prioritization, and the DASCD model for robust prediction. By leveraging attention mechanisms within a feed-forward convolutional network, DASCD captures complex non-linear interactions between clinical and biochemical features, outperforming conventional ML methods and standard DL architectures such as CNNs. This integrated framework not only achieves state-of-the-art predictive accuracy but also enhances interpretability and generalizability across heterogeneous datasets, effectively addressing the shortcomings identified in prior work.

Comparative analysis of recent PCOS detection methods and DASCD contributions is given in Table 1.

2.1. Motivation and problem statement

PCOS is a metabolic reproductive disorder in which the ovary produces abnormally large numbers of follicles. Radiologists always have difficulty detecting PCOS due to the variable follicle sizes and the disease's close association with veins and tissues. Pre-processing techniques are being utilized to extract meaningful data for analysis in order to forecast the PCO syndrome fertility and infertility for the data acquired from the KAGGLE repository. In addition to the patient's symptoms, doctors also need further tests and examinations to seek biochemical and clinical signs of PCOS. PCOS is diagnosed based on well-established clinical guidelines, most commonly the Rotterdam Criteria, which require at least two of the following three features including oligo/anovulation, clinical or biochemical hyperandrogenism, and polycystic ovarian morphology on ultrasound. International evidence-based recommendations guide the diagnostic process to ensure consistency and clinical accuracy. Gorilla Troops Optimization (GTO) is a nature-inspired algorithm that mimics the social and intelligent behavior of gorillas to efficiently explore and exploit the search space. In feature selection, GTO helps identify the most relevant features by balancing exploration and exploitation, reducing dimensionality, improving model accuracy, and enhancing generalization, especially

Table 1
Comparative analysis of recent PCOS detection methods and DASCD contributions.

Study (Year)	Approach (Accuracy)	Key Limitations	How Addressed in DASCD
Manjal et al. [22]	ExtraTree (88 %)	Small dataset, limited generalizability, no explicit feature prioritization	Robust feature selection + multi-dimensional attention improve generalization.
Bharati et al. [23]	Hybrid RF + Logistic Regression (91.01 %)	Limited interpretability, weak feature interaction analysis	SHAP + attention-based prioritization enhances interpretability
Chitra et al. [24]	Artificial Neural Network (92 %)	Black-box, risk of overfitting on small data	Lightweight design with squeeze layers reduces overfitting
Inan et al. [27]	XGBoost (95.83 %)	No explicit clinical feature importance	SHAP + attention for clinically relevant prioritization
Sayma Alam Suha et al. [17]	Stacking Ensemble (95.7 %)	Computationally heavy, low interpretability	Achieves comparable accuracy with a lighter interpretable framework
Rahman et al. [26]	RF + AdaBoost (94 %)	Relies on handcrafted features, lacks deep integration	Dense connectivity merges handcrafted + deep features
Proposed DASCD	Dense connections + squeeze layers + attention **(96 %+)**	–	Addresses dataset bias, interpretability, efficiency, and prioritization

useful in complex problems like PCOS detection. However, other studies used clinical information and text-based ultrasound reports rather than ultrasound images to identify PCOS. As a result, a DL strategy is used to simplify the training process and improve the detection accuracy.

One of the most significant challenges in PCOS detection is the inherent imbalance in the dataset, where the number of non-PCOS cases far exceeds the number of PCOS cases. This imbalance may cause the model to perform poorly as it could favor the majority class and could not identify the minority class correctly. A number of criteria, such as clinical characteristics, hormone levels, and ultrasound imaging, are involved in the diagnosis of PCOS. Feature selection is essential for high-dimensional datasets to remove redundant or unnecessary features, lower the risk of overfitting, and enhance the model's interpretability. Overfitting occurs when the model learns to memorize the training data, leading to poor generalization of unseen data. A model is said to be underfitting if it is unable to identify the underlying patterns in the data. Both issues can severely affect the accuracy and reliability of PCOS detection models. In order to overcome these obstacles, this study suggests a deep learning-based model that incorporates a number of modern techniques intended to improve the accuracy of PCOS identification.

3. Proposed methodology

One of the most common causes of female infertility is PCOS, which affects many women at reproductive age and can even persist well into their reproductive life. Therefore, this research develops an effective detection model with better performance. This research undergoes four essential phases: data acquisition and visualization, data pre-processing, feature selection, and detection. Pre-processing is crucial before processing the data in the following step. In order to ensure the greatest possible data description, this research entails finding the optimal feature subset(s). For an optimal feature selection, this research undergoes a hybrid bionic strategy, stated as GS^2TM. To strongly support detection performance, resample the data using a combination of SMOTE and ENN. This combination aims to solve the class imbalance issues and data outlier issues. The final stage is PCOS detection, and this can be done with a boosted deep learning model. The proposed detection model is based on densely connected layers, attention mechanisms, and squeeze convolution layers. The framework of the proposed methodology is illustrated in Fig. 1.

3.1. Data acquisition and pre-processing

The publicly accessible data set of PCOS patients from 'Kaggle' https://www.kaggle.com/datasets/prasoonkottarathil/polycysticovary-syndrome-pcos [21] is utilized as an input to the proposed PCOS detection model. This dataset contains the symptoms of patients together with the results of their PCOS diagnosis. Before using the dataset for training, it underwent a comprehensive investigation to gain a better understanding. The dataset comprises 541 records of female patients' data with 45 columns, encompassing various types of clinical data on PCOS abnormality. According to the primary study of the PCOS records, the dataset includes 541 records of female patients' data with 45 columns, including various forms of clinical information on the PCOS abnormality.

The proposed model is evaluated with a real-world benchmark PCOS detection dataset (https://www.kaggle.com/datasets/ankushpanday1/pcos-prediction-datasettop-75-countries/data), PCOS survey dataset (https://github.com/PCOS-Survey/PCOSData/), and Shreyas Vedpathak PCOS dataset (https://www.kaggle.com/datasets/shreyasvedpathak/pcos-dataset/data).

The UCI PCOS dataset, contributed by Kasturba Medical College, Manipal, India, contains clinically validated records of women with and without Polycystic Ovary Syndrome. It comprises 541 instances with approximately 44 attributes that cover demographic, clinical, biochemical, and ultrasound-based features. The dataset includes demographic details such as age, weight, height, BMI, and blood group; clinical and lifestyle indicators such as menstrual cycle regularity, cycle length, pregnancy status, history of abortions, and dietary and exercise habits; and biochemical markers including Follicle Stimulating Hormone (FSH), Luteinizing Hormone (LH), Anti-Müllerian Hormone (AMH), prolactin, thyroid-stimulating hormone (TSH), and derived ratios like LH/FSH. In addition, ultrasound features such as follicle count and average follicle size for both ovaries are provided. Unlike self-reported datasets, the UCI PCOS dataset is based on hospital records and laboratory investigations, ensuring higher biomedical reliability and making it suitable for evaluating computational models in a clinically meaningful context.

Before using DL models, the pre-processing stage is performed to address faults and anomalies in the datasets [32]. Therefore, replacing null values, balancing the dataset (SMOTE and ENN), and dropping unnecessary columns are employed in the pre-processing stage, and the steps are discussed below:

Replacing null values: Originally, the null values are processed during the data pre-processing phase. As they do not provide relevant information, features with excessive null or missing values are eliminated from the source. For instance, the feature "Unnamed" in the dataset has 539 null values; hence, it is removed. Features with few null values are replaced with more precise ones.

Dataset balancing (SMOTE and ENN): For data sampling, this research undergoes two sampling strategies, namely SMOTE and ENN, for data balancing and removing outliers.

To enhance the number of samples for the minority classes, the SMOTE model is introduced. Since the dataset has an uneven target attribute with 177 entries from people with PCOS and 364 records from people without PCOS, data balancing is performed to balance the distribution of classes for training. Using the SMOTE technique, a minority class's synthetic sample is created to balance out the desired attribute values. The mathematical formulation of the SMOTE method is shown in Eq. (1).

$$y_{sample} = y + \eta\,(y_{random} - y) \tag{1}$$

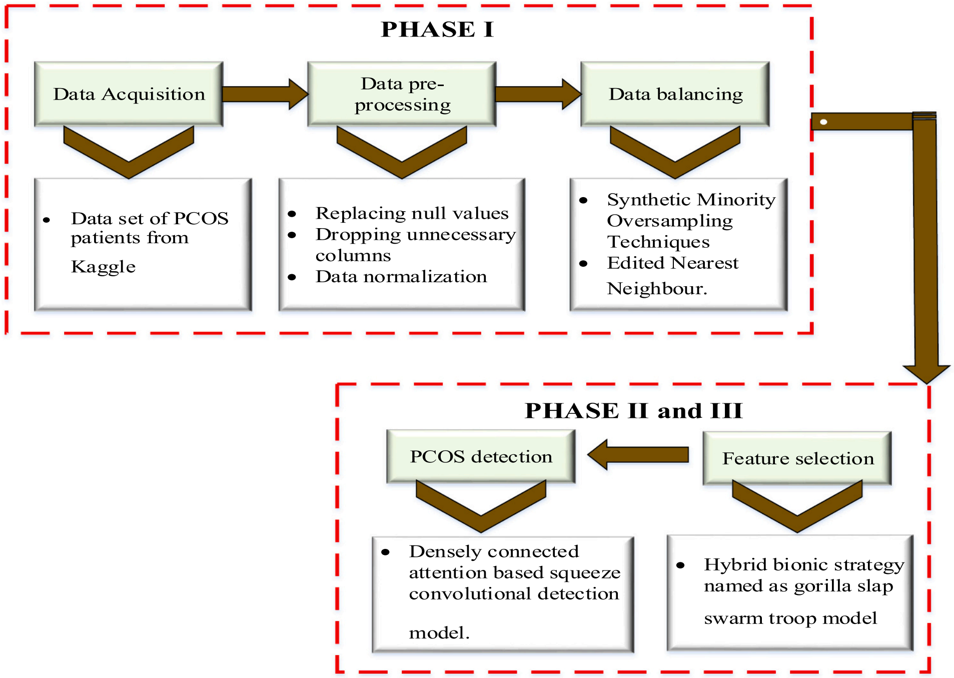

Fig. 1. Overall block diagram of the proposed methodology for PCOS detection, showing data acquisition, preprocessing, balancing, feature selection (GS2TM), and final detection model (DASCD).

here, y_{sample} defines the synthetic sample generated from the minority class value y and y_{random} defines the random sample chosen from the minority sample with $0 \leq \eta \leq 1$. The dataset instances in this study have grown to 728 records as a consequence of SMOTE, comprising 364 positive and negative PCOS diagnosis outcomes.

ENN is used to undersample the data after increasing the sample size and is also used for the majority of classes. However, it is utilized for all samples to exclude any instances of classes in this research. The major procedure for ENN is described below:

- Consider the sample$y \in N$. Here, N belongs to the entire sample count.
- Find K nearest neighbors for$y$.
- y could be eliminated if the percentage of neighbors from another class is high.
- The same procedure is repeated for each instance of the dataset.

In the healthcare industry, the problem of imbalanced datasets is well-known, especially for illnesses like PCOS, where the number of positive instances may be much smaller than the number of negative cases. Statistical models perform below optimally when this imbalance is not sufficiently addressed by standard data balancing procedures. In this approach, the SMOTE is employed in conjunction with ENN to balance the dataset and refine the decision bounds by eliminating noisy occurrences. For the minority class, SMOTE efficiently creates synthetic samples. At the same time, ENN contributes to cleaning the data by eliminating incorrectly classified occurrences, resulting in a more reliable and accurate model. This combination is highly effective for handling imbalanced data in the context of PCOS detection, where such imbalance is a common issue due to the complexity and variability of the condition.

Drop unnecessary columns and data normalization: The third phase in the pre-processing stage is to eliminate or drop unnecessary columns, and this process is essential to increase the detection accuracy. In the data set, one of two columns containing identical data is retained while the other is removed. For instance, the information provided by the "I beta-HCG (mIU/ML)" numerical column and the "II beta-HCG (mIU/ML)" categorical attribute is the same when "II beta-HCG (mIU/ML)" is omitted. The unnecessary columns "Sl. No." and "Patient File No." are also removed from the dataset because they simply include serial and patient file numbers, which may be ignored during future research. Data normalization is performed using the scalar MinMax approach, and the rescaling values of the features range from 0 to 1.

In this study, class imbalance, a common issue in PCOS datasets where the number of PCOS and non-PCOS samples is uneven, was explicitly addressed during the pre-processing phase. The approach involved a combination of SMOTE and ENN. SMOTE generates synthetic samples for the minority class by interpolating between existing minority class instances, increasing the representation of the underrepresented class and reducing bias toward the majority class. ENN is applied after oversampling to clean noisy or misclassified samples from both majority and minority classes. It removes instances that differ from the majority of their nearest neighbors, improving data quality and model robustness. Using SMOTE followed by ENN ensures that the dataset is not only balanced but also cleansed of noisy samples. This combined SMOTE-ENN strategy enhances the predictive performance of the model by allowing it to learn effectively from both classes without being skewed toward the majority class. By integrating this technique in preprocessing, the proposed pipeline effectively mitigates class imbalance, contributing to the high classification accuracy achieved by the DASCD model in distinguishing PCOS and non-PCOS cases.

3.2. Feature selection

Feature selection is the process of choosing a relevant subset of features from a high-dimensional dataset to improve model performance, reduce overfitting, lower computational cost, and enhance interpretability. In this study, a hybrid bionic strategy called GS2TM is employed for optimal feature selection. GS2TM is inspired by the social behavior of gorilla troops [34] and the dynamics of salp swarms. In gorilla troops, the dominant silverback leads the group in decision-making, foraging, and territory defense, which forms the basis for the exploration and exploitation phases of the algorithm. Similarly, salp swarm behavior models interactions and adjustments within the group, enhancing the search process and avoiding local optima.

GS2TM integrates these concepts as a swarm intelligence-based optimizer, where each agent (gorilla) represents a candidate feature subset. The optimizer searches a binary space (1 = feature selected, 0 = feature not selected) to identify the subset that maximizes classification performance while minimizing redundancy. Continuous position updates are converted to binary values using a transfer function to handle the combinatorial nature of feature selection. The strategy involves five key steps: fitness evaluation, control parameter adjustment, non-linear transfer function application, gorilla exploration, and position updates based on the S^2 model. By combining global exploration with local refinement, GS2TM efficiently identifies the most informative feature subsets, enhancing model accuracy with reduced complexity.

For searching the $N_S$dimensional space, the position vector $X$of each feature individual is specified in GS^2TM. The vector comprises a $N_S \times N_F$ dimensional matrix, as the position vector X consists of N_S feature individuals in the N_F dimensions. Therefore, the matrix can be represented below:

$$X = \begin{bmatrix} x_{1,1} & \cdots & x_{1,N_F} \\ \vdots & \ddots & \vdots \\ x_{N_S} & \cdots & x_{N_S N_F} \end{bmatrix} \tag{2}$$

In this section, the mathematical modeling of feature selection is described. In general, a dataset has $N_S \times N_F$ dimensions, where $N_S$stands for the overall sample count and N_F specifies the overall feature count. The fitness function is described in the equation below:

$$Fitness = \gamma \times \lambda s + \left(1 - \gamma\right) \times \left(\frac{|S|}{N_F}\right) \tag{3}$$

here, λs specifies the detection error, $\gamma$is the regularization factor, and $|S|$ specifies the selected features. To maintain the balance between $\left(\frac{|S|}{N_F}\right)$and $\lambda s, \gamma$ is applied. Exploration and exploitation are balanced when the randomization parameters are carefully adjusted. The equation is given below:

$$CR = 2 \times rand - 1 \tag{4}$$

The transfer function governs the balance between exploration and exploitation. Its non-linear formulation is given in Eq. (5).

$$NF = \text{Sin}\left(\phi - \frac{it}{Max_{It}}\right) \tag{5}$$

The term ϕ represents the constant term, it and Max_{It} specify the current and maximum iterations, respectively.

The exploration stage of the GT model has three phases, and the mathematical expression is given below:

$$GA\left(t+1\right) = \begin{cases} (Upper_{Band} - Lower_{Band}) \times r_1 + Lower_{Band}, & rand < P \\ (r_2 - C) \times A_r(t) + L \times H, & rand \geq 0.5 \\ A(i) - L \times (L \times (A(t) - GA_r(t)) + r_3 \times (A(t) - GA_r(t))) & rand < 0.5 \end{cases} \tag{6}$$

here$GA(t+1)$ and $A(t)$ represent the position vectors of features in the next and current iteration. In each iteration, the random values r_1, r_2 and r_3 ranges from 0 to 1. The position vector C, L, H are computed as follows. Eq. (6) shows the original three-phase logic of the GTO exploration stage, while Eq. (7) provides a unified expression combining all phases into a single continuous update. The proposed model adjusts the control parameter and non-linear function by replacing the random parameter from Eq. (6), as shown in Eq. (7).

$$GA(t+1) = A(t) - CR \times NF \times L \times (L \times (A(t) - GA_r(t)) + (A(t) - GA_r(t))) \tag{7}$$

The exploitation stage is performed using salp swarm optimization. In the gorilla troop model it consists of two phases: observing the silverback and competing for mature females. The first phase, observing the silverback, involves the leader making decisions, managing the troop, and guiding it to resources, as shown below:

$$GA(t+1) = L \times M \times (A(t) - A_{silverback}) + A(t) \tag{8}$$

The gorilla's position vector is represented as $A(t)$ and $A_{silverback}$ defines the position vector of a silverback gorilla. The value of M is defined as follows.

$$M = \left(\left|\frac{1}{N}\sum_{t=1}^{N} GX_i(t)\right|^g\right)^{\frac{1}{g}} \tag{9}$$

$$g = 2^L \tag{10}$$

where N is the total gorilla, $GX_i(t)$ is the position vector of each gorilla. The randomization control parameter, missing in the above equation, is incorporated in the updated form of Eq. (8) as follows:

$$GA(t+1) = CR \times NF \times L \times |A(t) - A_{silverback}(t)| + A(t) \tag{11}$$

here, NF represents the non-linear transfer function that guides the agents progressively in the direction of the global optimal position. The second stage of the exploitation phase is competing for mature females, and the mathematical expression is given below:

$$GA(i) = A_{silverback} - (A_{silverback} \times Q - A(t) \times Q) \times Y \tag{12}$$

It is clear from this equation that the agents follow any unknown location obtained by multiplying the ideal position by the variable Q that may affect the convergence speed. Therefore, the mathematical expression of the updated exploration phase is shown in Eq. (11).

$$X_{i,j} = LB_j + rand \times (UB_j - LB_j) \tag{13}$$

Eq. (13) is an updated version of the original exploration phase logic (Eq. 6), introducing adaptive scaling based on population diversity and integration with the Crossover Rate (CR) and Normalization Factor (NF) to better control exploration intensity. This formulation focuses solely on the exploration phase, enhancing computational efficiency and maintaining multi-strategy behavior. The goal is to make the GS2TM algorithm more adaptable and diverse in searching the solution space, enabling faster identification of optimal feature subsets. Fig. 2 illustrates the flow chart of the proposed feature selection model. GS2TM is particularly suited for high-dimensional medical datasets, such as those used in PCOS identification, because it combines exploitation of prominent features with exploratory swarm intelligence, efficiently removing redundant or irrelevant features. By selecting the most significant features, this approach improves both the accuracy and interpretability of the model.

3.3. PCOS detection

The deep learning model plays a major role in the medical domain

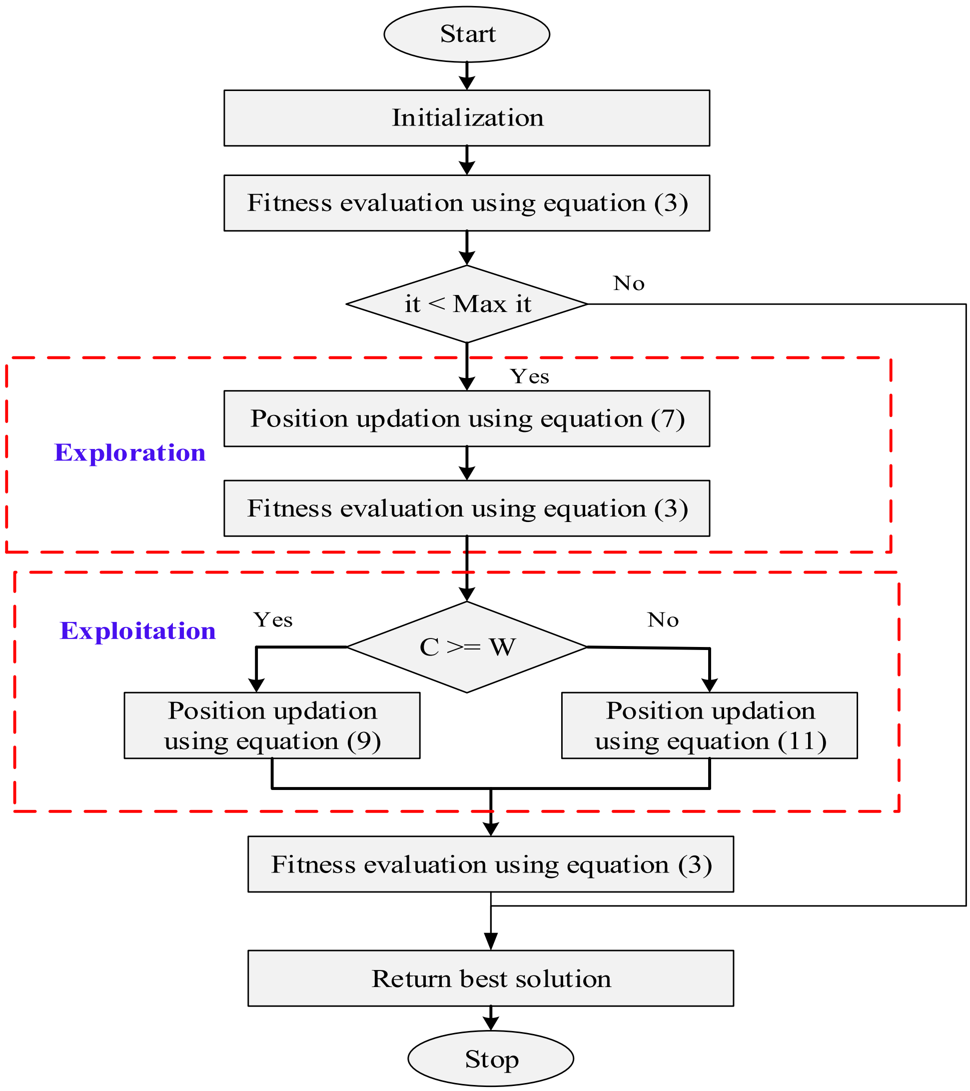

Fig. 2. Flowchart of the feature selection process, highlighting key steps to identify the most relevant features for PCOS detection.

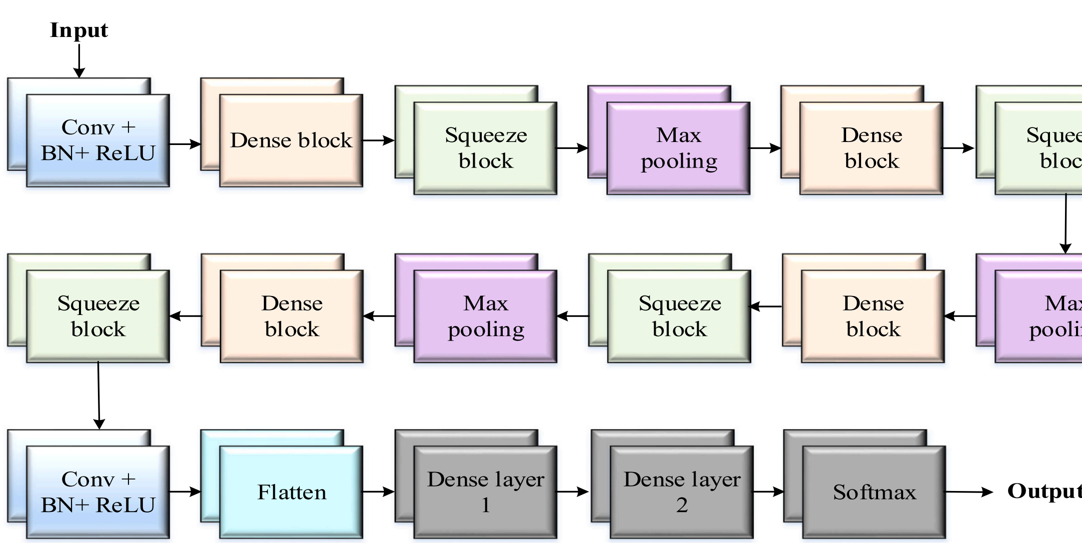

Fig. 3. Network architecture of the proposed PCOS detection model showing the layers and connections used for feature-based classification.

due to its high, fast, and accurate detection accuracy. Based on the significance of the deep learning model, the DASCD model is proposed for PCOS detection. The proposed network model is emphasized by several mechanisms, including dense connectivity, squeeze, and attention mechanisms. The proposed model consists of several convolutional layers, a batch normalization layer, pooling layers, flattened layers, a dense layer, and classification layer, and a sigmoid activation function. Also, the proposed architecture insists on four dense, squeeze, and attention mechanisms with various convolution layers. To prevent overfitting, the DASCD model incorporates attention mechanisms that focus on the most informative features while ignoring irrelevant or noisy inputs. Its densely connected architecture promotes feature reuse and ensures effective gradient flow. Additionally, feature selection via GS2TM reduces the input to the most discriminative 23 attributes, and SMOTE + ENN balancing generates representative samples for minority classes, enabling the model to learn generalizable patterns. The network architecture of the PCOS detection model is illustrated in Fig. 3. The attention technique is used before the flatten layer.

3.3.1. Attention mechanism

Weighted summation, which modifies weights based on the importance of information using various similarity measurement approaches, is the central component of the attention process. Feature extraction aims to extract features essential for PCOS detection. Therefore, the final refined feature output can be represented in Eq. (14).

$$F' = F \otimes Att(F) \tag{14}$$

here, $Att(.)$ defines the attention block calculation, which is used to derive the attention map and $\otimes$ specifies the element-wise multiplication. Attention values are transmitted along many axes during multiplication. The attention mechanism can be emphasized with two different attention blocks, namely, temporal and channel attention blocks.

Temporal attention block: The output of the convolutional block is defined as a feature map, and the mathematical representation is defined as $F \in R^{T\times W}$. Here, T and Wstands for the length and quantity of the feature maps, respectively. In order to create two channel context descriptors and calculate temporal attention, max-pooling and average-pooling procedures are first applied along the channel axis $F^t_{\max} \in R^{1\times T}$ and $F^t_{avg} \in R^{1\times T}$. Therefore, the computation process is defined in the equation below:

$$F^t_{\max} = \max(F(1), F(2), \ldots F(L)) \tag{15}$$

$$F^t_{avg} = \frac{1}{T}\sum_{i=1}^{T} F(i) \tag{16}$$

To create the temporal attention map$Att(F) \in R^{1\times T}$, the attention maps are combined through a 1-D convolution with a kernel size of nine. Therefore, the mathematical expression is given below:

$$Att(F) = \sigma\left(Conv^9\left(F^t_{\max} \otimes F^t_{avg}\right)\right) \tag{17}$$

here, σspecifies the sigmoid activation function. The output F'_t is examined by multiplying the activated weights in an element-wise format using the initial feature map$F$.

$$F_t{}' = F \otimes Att_t(F) \tag{18}$$

Channel attention block: Based on the examination of the previous attention block, initially, the max pooling $F^c_{\max} \in R^{1\times W}$ and average pooling feature maps $F^c_{avg} \in R^{1\times W}$are extracted. Then, the feature maps are fed as an input to the fully connected FClayer with N number of units added element wise. The original feature map is multiplied by the weights, and the channel attention block computation formula is described below:

$$F'_c = F \otimes \sigma\left(FC^N\left(F^c_{\max} + F^c_{avg}\right)\right) \tag{19}$$

The output of the temporal attention block is a vector of attention weights over the time steps (or spatial/feature positions). Before applying element-wise multiplication with the feature map, the attention vector is reshaped and broadcast across the feature dimension to ensure dimensional compatibility. Specifically, the attention weights of shape ($B \times T \times 1$)(B×T × 1) are multiplied with the feature map ($B \times T \times D$)(B×T × D), where broadcasting is used along the last dimension. The attention mechanism produces a weight vector over the temporal dimension. The attention output is extended or reshaped and propagated to match the shape of the feature map before element-wise multiplication. This ensures that each attention weight is applied consistently across the relevant dimension of the feature map.

3.3.1.1. *Dense connectivity.*

Dense connections help address the vanishing gradient problem and promote feature reuse by connecting each layer to all subsequent layers. This design allows the network to directly pass information and gradients between layers, enhancing learning efficiency. The convolutional layers consist of multiple filters of varying sizes to capture diverse features from the input data. To reduce computational complexity and retain the most important information, pooling layers are used to extract dominant features. The flattening layer then converts the resulting 2D feature maps into a 1D array, which is passed to the fully connected layer. The fully connected layer combines the features learned from previous layers to produce the final output. This approach reduces the total number of parameters and improves overall learning efficiency. In the DASCD model, each layer receives feature maps from all preceding layers, and its mathematical formulation is provided below:

$$x_1 = H_1([x_0, x_1, \ldots, x_{D-1}]) \tag{20}$$

here, H_l specifies the transformation and $[x_0, x_1, \ldots, x_{D-1}]$ defines the concatenated output of all previous layers. The concatenation enriches the feature representation, ensuring that the model learns both low- and high-level abstractions simultaneously.

3.3.1.2. *Squeeze convolution layers.*

SqueezeNet is a type of CNN with a more compact design while maintaining high accuracy. The model's computational footprint is reduced through the squeeze operation (1 ×1 convolutions), which lowers the dimensionality of the input feature maps. These feature maps are then processed to extract deeper spatial information via the expand operation (1 ×1 and 3 ×3 convolutions). This combination keeps the network lightweight while capturing the essential spatial features required for PCOS diagnosis. The squeeze and expand layers work together to capture both local and global patterns, which are critical for identifying subtle abnormalities associated with PCOS. The architecture of the dense and squeeze blocks is illustrated in Fig. 4.

The proposed framework is designed for practical integration into clinical diagnostic workflows for PCOS. The system takes patient data such as clinical, hormonal, and ultrasound features as input. In the preprocessing phase, missing values are handled, redundant columns are removed, and class imbalance is corrected using the SMOTE-ENN hybrid method to ensure reliable model learning. In the feature selection phase, the proposed GS2TM identifies the most informative features, effectively reducing the feature dimension by 51.1 % and retaining only the dominant clinical predictors. Finally, in the detection phase, the DASCD classifies patients into PCOS and non-PCOS categories based on the selected features.

This end-to-end process can be employed as a decision-support tool for clinicians. By emphasizing feature prioritization and explainable attention mechanisms, the model highlights influential clinical parameters contributing to PCOS diagnosis, thus enhancing transparency and

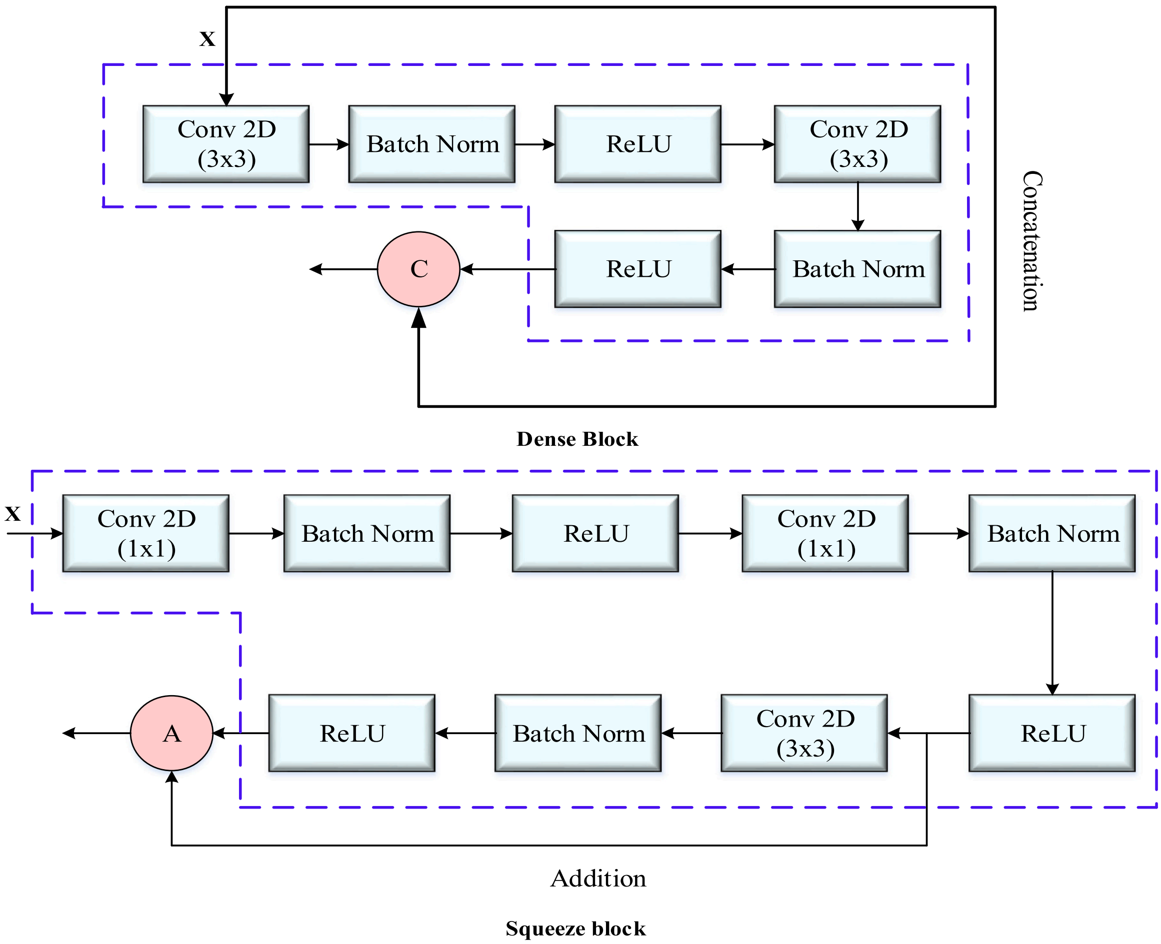

Fig. 4. Architecture of the dense and squeeze block illustrating the layer connections and feature extraction flow.

supporting medical experts in early and accurate disease detection. The framework can be integrated into electronic medical systems or deployed as a screening tool for gynecologists and endocrinologists.

3.4. *Justification of the proposed network*

The superior performance of the proposed hybrid PCOS detection framework can be mathematically justified by analyzing each stage: data balancing, feature optimization, and classification network design.

3.4.1. *SMOTE–ENN for data balancing*

Let the original dataset be represented as

$$D = \{(x_i,\ y_i)\}_{i=1}^{N}, \qquad x_i \in \Re^d,\ \ y_i \in \{0,\ 1\} \tag{21}$$

where$y_i = 1$ denotes PCOS and $y_i = 0$ denotes non-PCOS. Due to class imbalance $(N_+ << N_-)$, the minority class distribution $p(x|y = 1)$ is poorly estimated.

SMOTE generates synthetic minority instances as

$$x_{new} = x_i + \lambda(x_m - x_i), \qquad \lambda \sim U(0,\ 1) \tag{22}$$

where x_{nn} is the k-nearest neighbor of x_i. This process increases the support of the minority class and reduces the variance of its sample distribution, improving boundary learning.

ENN subsequently removes noisy or mislabeled samples: $x_j \in D$ is removed if the majority (k-NN$((x_j)) \neq y_j$. Thus, the combined SMOTE–ENN mechanism smoothens the decision boundary and reduces overlapping regions between classes. The overall effect can be approximated by reducing the empirical risk variance term in the generalization error bound:

$$E[R(h)] - R_{emp}(h) \infty O\left(\sqrt{\frac{VC(H)}{N_{eff}}}\right) \tag{23}$$

where, N_{eff} is effectively increased by balancing and denoising, improving classifier generalization.

3.4.2. *GS2TM for optimal feature selection*

Let $z = [z_1, z_2, \ldots . z_d] \in \{0,\ 1\}^d$ be a binary vector indicating whether feature j is selected $z_j = 1$.

The optimization problem is defined as:

$$\max F(z) = \alpha . Acc_{val}(z) - \beta . \frac{\|z\|_1}{d} \tag{24}$$

whereα *and* β control the trade-off between accuracy and feature subset size.

The Gorilla–Salp–Swarm–Troop Model (GS2TM) hybridizes exploration and exploitation strategies:

- ➢ Gorilla exploration: random long jumps enhance global search.
- ➢ Salp chain update: smooth leader–follower propagation encourages convergence.
- ➢ Troop local search: greedy one-bit flips ensure local refinement.

Through these cooperative operators, GS2TM minimizes feature redundancy while retaining the most discriminative attributes. Reducing dimensionality from d *to* m (Where $m << d$) decreases the model capacity and improves generalization.

Given that the generalization error bound depends on VC-dimension($VC(H) \infty m$):

$$R(h) \leq R_{emp}(h) + O\left(\sqrt{\frac{m}{N}}\right) \tag{25}$$

The reduction in m achieved by GS2TM tightens the bound, thus improving reliability and reducing overfitting.

3.4.3. DASCD: densely connected attention-based squeeze convolutional model

The DASCD classifier integrates dense connectivity, squeeze–expand convolutions, and attention mechanisms for robust prediction.

Each dense block computes:

$$X_l = H_l([X_0,\ X_1,\ \ldots, X_{l-1}]) \tag{26}$$

where $H_l(.)$ is the composite function of Batch Normalization, ReLU, and Convolution. Dense connections improve gradient flow and enable feature reuse, mitigating vanishing gradients and overfitting.

The squeeze-expand module reduces dimensionality:

$$U_{squeeze} = \varphi(W_{1*1}\ * U), \tag{27}$$

$$U_{\text{expand}} = \varphi\left(W_{3*3} * U_{squeeze}\right), \tag{28}$$

Thereby decreasing parameters and regularizing the network. The attention mechanism adaptively reweights channels:

$$a = \sigma(W_2\delta(W_1 s)), s_c = \frac{1}{HW}\sum\nolimits_{i,j} F_{c,i,j},\ \ F_c^{'} = a_c.F_c \tag{29}$$

where s represents global pooled descriptors, and $a_c$emphasizes informative channels.

This selective amplification increases the signal-to-noise ratio of relevant features, enhancing decision reliability.

The total loss function is:

$$L(\theta) = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log p_\theta(x_i) + (1-y_i)\log(1-p_\theta(x_i))] + \lambda\|\theta\|_2^2 \tag{30}$$

3.4.4. Proof of enhanced performance over existing models

The combined mathematical mechanisms yield measurable improvements:

- ➢ SMOTE–ENN increases the effective training size $N_{eff}$and reduces class imbalance noise, lowering empirical risk variance.
- ➢ GS2TM minimizes redundancy and reduces VC-dimension, leading to tighter generalization bounds and less overfitting.
- ➢ DASCD introduces dense gradients, bottleneck regularization, and adaptive attention weighting, improving representational power with fewer parameters.
- ➢ The unified pipeline aligns bias–variance trade-offs across stages:

$$TotalError = Bias^2 + Variance + Noise \tag{31}$$

SMOTE–ENN reduces data noise, GS2TM lowers variance, and DASCD reduces bias through deep contextual learning. Empirically, the proposed model achieved higher accuracy using only 23 features, surpassing conventional ML and deep learning models. This performance gain is theoretically consistent with reduced model capacity, improved feature discriminability, and enhanced generalization derived from the above formulations.

4. Experimental setup

4.1. Dataset description

The proposed PCOS detection model was evaluated using the publicly available Kaggle PCOS dataset, along with additional benchmark datasets (Real-world benchmark, PCOS survey, Vedpathak PCOS, and UCI PCOS) for external validation. These datasets contain clinical, biochemical, and demographic information of patients, with labels indicating PCOS or non-PCOS conditions. Since medical datasets often suffer from imbalanced class distributions, the initial dataset analysis revealed a higher number of non-PCOS samples compared to PCOS cases.

4.2. Preprocessing and data balancing

In the preprocessing stage, missing values were replaced, irrelevant columns were removed, and numerical features were normalized for stable training. To overcome the issue of class imbalance, a hybrid resampling strategy combining Synthetic Minority Oversampling Technique (SMOTE) and Edited Nearest Neighbor (ENN) was applied. SMOTE generates synthetic minority class examples, while ENN removes noisy and borderline samples, resulting in a cleaner and more representative dataset. This balancing step ensures that the model learns equally from both PCOS and non-PCOS cases.

4.3. Feature selection using GS2TM

To reduce redundancy and improve model efficiency, the Gorilla Salp Swarm Troop Model (GS2TM) was employed for feature selection. GS2TM, a hybrid bio-inspired optimization method, was used to identify the most discriminative attributes contributing to PCOS prediction. This process reduced the feature set by 51.1 %, retaining only 23 optimal features out of the original dataset. Feature selection not only minimized computational complexity but also enhanced the model's ability to focus on clinically relevant factors.

4.4. Proposed detection model (DASCD)

The Densely Connected Attention-Based Squeeze Convolutional Detection (DASCD) model was developed for PCOS detection. This architecture integrates dense connections, which promote feature reuse and effective gradient propagation, with attention mechanisms that highlight the most relevant features for classification. Multiple convolutional and fully connected layers were organized in a feed-forward manner. Hyperparameters were optimized using trial experiments, with the final configuration including: batch size = 32, learning rate = 0.001, optimizer = Adam, and maximum epochs = 300.

4.5. Training and validation strategy

To ensure robustness and generalizability, several validation strategies were employed:

10-fold cross-validation: The dataset was partitioned into ten folds, with the model trained on nine and validated on the remaining fold, repeated ten times.

Nested cross-validation: Used for hyperparameter tuning and unbiased performance estimation.

Hold-out testing: Independent datasets were used to further evaluate the model's generalization ability across different clinical sources.

4.6. Dataset splitting and experimental design

After sampling, there are 724 occurrences in the dataset, including 362 positive and negative PCOS diagnoses in the Kaggle dataset. For model development, the UCI and Kaggle datasets were split into training, validation, and testing sets using a 70:20:10 ratio, resulting in 379 records for training, 81 for validation, and 81 for testing in each dataset. The PCOS Survey dataset and Shreyas Vedpathak dataset were used entirely for external evaluation to assess model generalizability. A similar split has been used for existing comparison approaches.

4.7. Simulation parameters

The hyperparameters used in the proposed approach are summarized below. For SMOTE, five K-neighbors were used with an auto-sampling strategy and a random state of 42. The parameter η was set to 0.6. For ENN, three neighbors were considered, and misclassified majority class examples were removed. This integration was applied after null value replacement and feature cleaning to improve class balance and minimize minority class noise. The elimination criterion followed the standard ENN rule: a majority class sample was removed if its class label differed from the majority of its K neighbors. This process effectively eliminated borderline and noisy samples, which are especially problematic in imbalanced datasets. No additional thresholding beyond the ENN rule was applied.

Null value replacement was performed using median imputation for all numerical columns. Median values were calculated from the non-missing entries of each feature and used to fill missing values. Median imputation was chosen to reduce the influence of outliers, which are common in clinical datasets. For the GS2TM model, the simulation parameters were as follows: population size = 20, number of epochs = 300, crossover rate = 0.8, mutation rate = 0.1, detection error (λs) = 0.013, regularization factor (γ) = 0.01, constant (φ) = 5, upper bound = 1, lower bound = 0, parameter η = 0.6, and K = 3. For DASCD, the simulation included 4 convolution blocks, kernel size = 3, sigmoid activation function, batch normalization, a self-attention module, and a dropout rate of 0.4.

4.8. Evaluation metrics

The performance of the proposed model was assessed using widely accepted metrics in medical diagnostics: Accuracy, Precision, Recall, F1-score, and Area Under the ROC Curve (AUC). These metrics provide a comprehensive assessment of diagnostic ability, balancing correct classification, sensitivity to PCOS cases, and overall reliability.

The performance metrics used in this study are listed as follows.

$$\text{Accuracy} = \frac{T_P + T_N}{T_P + F_P + F_N + T_N} \tag{32}$$

$$\text{Precision} = \frac{T_P}{T_P + F_P} \tag{33}$$

$$\text{Recall} = \frac{T_P}{T_P + F_N} \tag{34}$$

$$\text{F1} - \text{score} = \frac{2.\ Precision.\ Recall}{Precision + Recall} \tag{35}$$

$$\text{AUC} = \frac{T_P\ rate + T_N rate}{2} \tag{36}$$

where, T_P represents true positive, T_N represents true negative, F_P represents false positive, and F_N represents false negatives.

5. Results and discussions

5.1. Impact of data preprocessing and balancing

The major focus of the pre-processing stage is data balancing, which is executed with the combination of SMOTE and ENN balancing models. Also, in the pre-processing stage, the columns such as serial number, patient file number, and unnamed columns are removed. The proposed data sampling model is compared with the existing classifier models without adjusting the sampling phase. Therefore, the tenfold cross-validation stage is evaluated with and without applying the SMOTE. The overall cross-validation score with and without SMOTE is illustrated in Fig. 5. The figure shows that the proposed model attains better detection accuracy by including a sampling method. As shown in the figure, the detection accuracy across folds is consistently higher when SMOTE and ENN are applied, clearly demonstrating the positive impact of balancing on model robustness. The height of the bars indicates that the balanced dataset not only improves average accuracy but also reduces variation across folds, reflecting greater stability in performance. The performance improved because SMOTE increased the representation of minority PCOS cases, allowing the model to learn balanced decision boundaries. ENN further refined the dataset by removing noisy and overlapping samples. Together, they enhanced class separability, reduced bias toward the majority class, and improved the overall cross-validation scores compared to the imbalanced dataset.

5.2. Feature selection analysis

The second foremost stage is feature selection; in this stage, the hybrid bionic model is introduced. A parametric statistical test called Analysis of Variance (ANOVA) can be used to assess the means of two or more samples of data coming from the same distribution. Therefore, the ANOVA test statistic can be evaluated using the following equation:

$$F = \frac{\sum n_i\,(\overline{x_i} - \overline{x})^2/(k-1)}{\sum\sum\,(\overline{x} - \overline{x_i})^2/(N-k)} \tag{37}$$

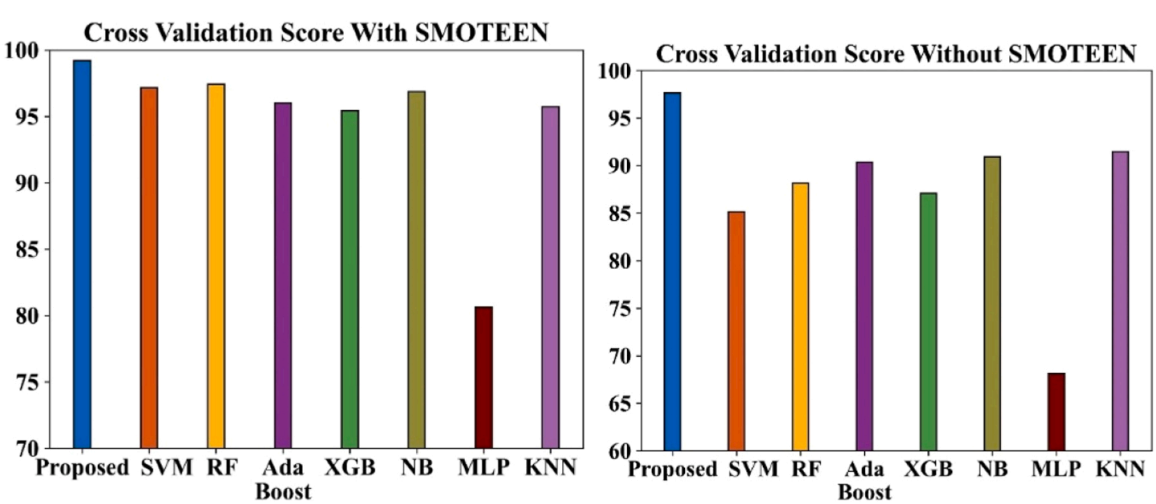

Fig. 5. Cross-validation scores of the PCOS detection model with and without using SMOTE and ENN for data balancing.

Table 2
Dominant selected features and their statistical analysis.

Features	P- values	Features	P values
Age (yrs)	1.1022e-07,	PRL(ng/ML)	0.015
Weight	2.8e-09,	PRG(ng/ML)	0.029
BMI	2.17e-08,	RBS(mg/dl)	0.031
Blood Group	0.004	Weight gain(Y/N)	2.618e-43,
Pregnant(Y/N)	0.030	hair growth(Y/N)	9.259e-26,
No. of abortions	0.014	Fast food (Y/N)	3.055e-33,
FSH(mIU/ML)	0.311	Reg.Exercise(Y/N)	1.394e-07,
FSH/LH	0.24	Follicle No. (L)	4.634e-38,
Hip(inch)	3.3376e-08	Follicle No. (R)	1.006e-56,
Waist(inch)	2.367e-06	Avg. F size (L) (mm)	0.00037
Waist:Hip Ratio	0.161	Endometrium (mm)	0.0026
AMH(ng/ML)	8.572e-16		

here, n_i represents the sample size, $\overline{x_i}$ specifies the sample mean, and $\overline{x}$ defines the overall mean. The total number of observations used in the analysis, and the number of independent groups are denoted by$N$and k, respectively. The statistics from the ANOVA test are then matched with the corresponding P-values, and for this test, the threshold for P-values is set at 0.05. The proposed model uses the GS2TM for feature selection, which reduces the feature set from 45 to 23 features, effectively selecting 51.1 % of the original features. This notable decrease shows how GS2TM may remove superfluous and unnecessary features, improving the model's effectiveness and readability.

Features like "Sl. No.," "Patient File No.," and highly correlated columns (such as "Beta HCG I" and "Beta HCG II") are eliminated during preprocessing because they are redundant or irrelevant. From these two columns, II beta-HCG (mIU/ML) is retained. This improves the generalization of the model and decreases noise. Shapley Additive Explanations (SHAP) is used to further validate the significance of each attribute chosen. To ensure non-redundancy, the SHAP summary plot (to be provided as a figure) verifies each feature's distinct contribution to the prediction model. The statistical analysis is shown in Table 2.

The extracted features are represented in Fig. 6. These feature details are utilized by the proposed approach for the evaluation of PCOS diagnosis. PCOS features were identified through the novel hybrid feature selection algorithm, GS2TM, which prioritizes features based on their relevance and statistical contribution to PCOS diagnosis. Hence, these features are most informative for the detection process.

5.3. Detection performance analysis

5.3.1. Training loss and accuracy

The accuracy and loss curve analysis of the proposed PCOS detection model are presented in Fig. 7. The model's performance was evaluated across training, testing, and validation phases by varying the number of epochs. The bar chart of accuracy across epochs shows a clear upward trend in the initial phase, with accuracy improving steadily and surpassing 90 % by the 50th epoch. Between epochs 200 and 300, the curve stabilizes, demonstrating that the model achieves consistent generalization without signs of overfitting. Similarly, the loss curve exhibits a monotonic decline for both training and validation sets, converging

Table 3
Comparative analysis with (23 features 51.1 %) and without feature selection.

Methods	Accuracy (%)	Accuracy (%)
LR	87.42	87.91
SVM	87.48	87.30
DT	81.10	82.63
RF	89.57	90.18
NB	81.90	80.85
AdaB	85.64	85.89
Proposed	97.6	99.2

```
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Fig. 6. Features selected by the feature selection process for use in PCOS detection.

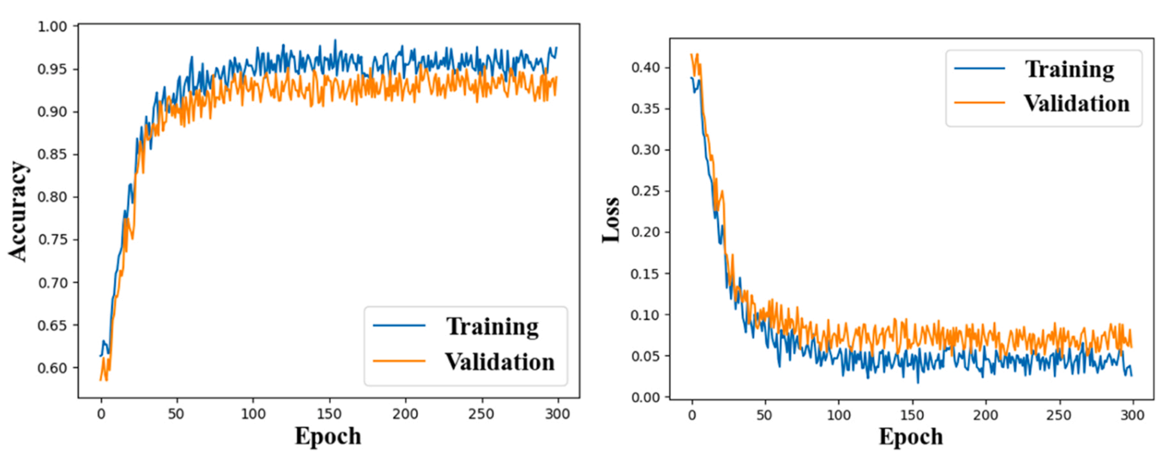

Fig. 7. Accuracy and loss curve of the proposed PCOS detection model across training and validation iterations.

smoothly after 250 epochs, which confirms the robustness and stability of the learning process. The close alignment of training and validation curves further demonstrates that the model effectively learns discriminative patterns while maintaining robustness. The improved performance is mainly attributed to the integration of data balancing, effective feature prioritization, and the hybrid learning framework with attention mechanisms, which together enhance class separability, reduce bias, and allow the model to converge faster with higher accuracy and lower loss.

5.3.2. Performance metrics

In this section, several evaluation metrics such as accuracy, precision, recall, F1-score, and AUC are analyzed with and without the feature selection strategy. The proposed model is also compared with existing machine learning classifiers to demonstrate its efficacy for PCOS detection. Accuracy represents the proportion of correctly classified predictions to total predictions. Precision denotes the proportion of true positive predictions among all predicted positives. Recall measures the percentage of actual positives correctly identified by the model. The F1-score is the harmonic mean of precision and recall, offering a balanced assessment of performance. The comparative analysis under these metrics, with and without feature selection, is shown in Table 3.

Among the baseline models, logistic regression (LR) employs a sigmoid function for binary classification with reasonable efficiency but is limited to complex, non-linear data. The adaptive boosting (AdaBoost) classifier is powerful yet highly sensitive to noisy or poor-quality data. Support Vector Machines (SVM) perform well in multiclass classification but require careful kernel selection and significant training time. Decision Trees (DT) are simple but unstable, as minor changes in data can drastically alter the tree structure, reducing accuracy. Random Forest (RF), an ensemble method, mitigates overfitting and manages high-dimensional data effectively, but at the cost of longer training times. Naïve Bayes (NB) is efficient for small datasets but performs poorly with large, complex data. Compared to these traditional classifiers, the proposed deep learning–based DASCD model demonstrates superior performance both with and without feature selection. Applying the GS2TM feature selection strategy yields notable improvements across all performance metrics: recall improves from 96.54 % to 98.8 %, accuracy from 96.3 % to 98.7 %, and precision from 97 % to 99 %. While the numerical improvements appear marginal, they are significant in clinical decision-making contexts.

The improvements can be attributed to three major factors:

1. Feature Redundancy Reduction: The original dataset contains 45 features, many of which are redundant or noisy. GS2TM effectively prioritizes 23 relevant features, reducing overfitting risk and enhancing model generalization.
2. Cross-Dataset Generalization: By eliminating irrelevant features, the model achieves robustness across multiple datasets. External validation confirms that the reduced feature set consistently performs well in unseen, real-world cases.
3. Computational Efficiency: Selecting an optimal subset of features lowers the training cost of the deep learning model, which is crucial for deployment in resource-constrained healthcare environments.

Additionally, the potential integration of reinforcement learning (RL) into the feature selection stage is discussed. While GS2TM offers strong global optimization through nature-inspired metaheuristics, RL could provide adaptive, reward-driven selection. However, RL introduces additional complexity, including careful reward design, high computational cost, and tuning requirements, which may hinder real-time clinical applicability. Hence, GS2TM alone provides an effective balance between performance and efficiency.

5.3.3. ROC and confusion matrix

The analysis of the Receiver Operating Characteristic (ROC) curve and confusion matrix is presented in this section. Fig. 8 depicts the confusion matrices of the proposed detection model with and without feature selection. The matrix visualization clearly demonstrates that the inclusion of feature selection reduces the number of misclassified cases. With feature selection, 64 out of 67 diseased cases are correctly identified, and only 3 are incorrectly predicted as non-diseased, while without feature selection, 62 are correctly classified and 5 are misclassified. This qualitative observation highlights that the decision boundaries become more refined when redundant features are eliminated, leading to sharper class separation. These findings quantitatively and visually demonstrate the benefit of the feature selection stage. The performance enhancement is primarily due to the feature selection stage, which eliminates irrelevant or redundant attributes, reduces noise, and highlights the most discriminative features for classification. This allows the model to focus on clinically significant predictors, thereby improving classification accuracy and reducing misclassification errors.

Fig. 9 shows the ROC analysis for both cases. The ROC curve with feature selection consistently lies above the curve without feature selection across all thresholds, indicating better trade-offs between sensitivity and specificity. The area under the ROC curve (AUC) further quantifies this improvement, confirming the robustness and reliability of the proposed model. The visual separation between the two ROC curves qualitatively illustrates that the model with feature selection maintains stronger discrimination ability, even in borderline cases.

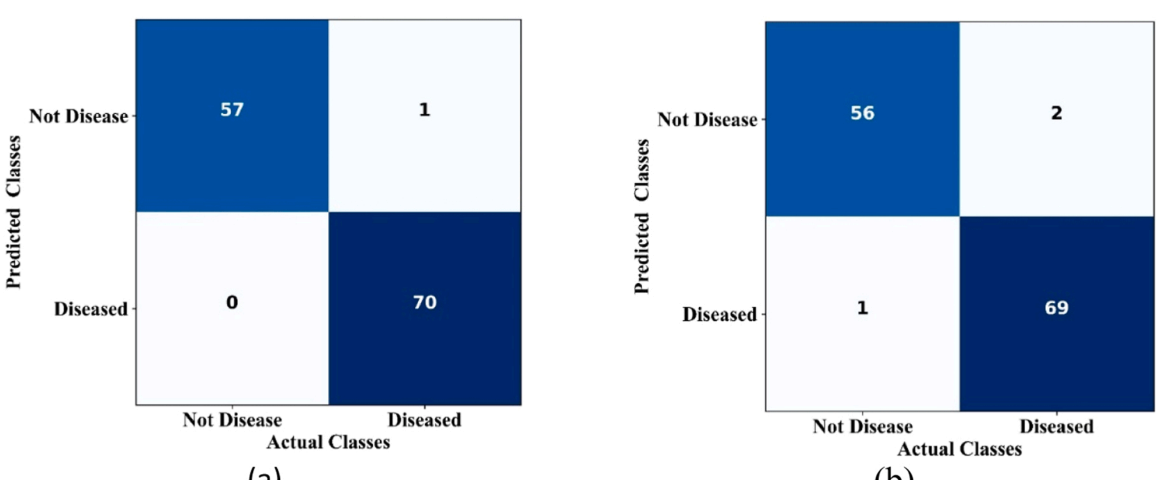

Fig. 8. Confusion matrices of the PCOS detection model comparing performance (a) with feature selection and (b) without feature selection.

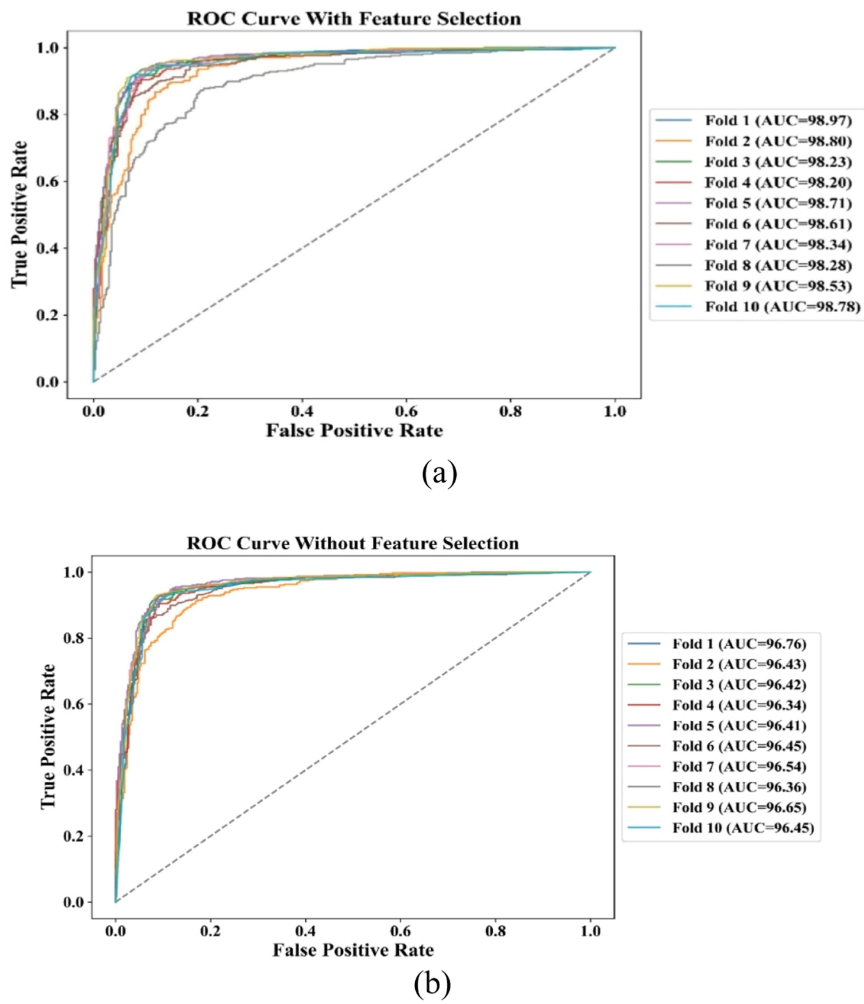

Fig. 9. ROC curves for the proposed PCOS detection model using 10-fold cross-validation (a) With feature selection (b) Without feature selection.

5.4. Model explainability (SHAP analysis)

A Shapley Additive Explanations (SHAP) analysis [29] was conducted to enhance the interpretability of the proposed framework. The final DASCD model trained on the 23 features selected by GS2TM was integrated into the SHAP framework to compute feature contributions. Fig. 10 illustrates the SHAP dependence plots, which highlight how specific variables influence PCOS classification. The SHAP analysis highlights follicle count, hormone levels, and BMI as the most influential features in PCOS prediction. Clinically, these attributes correspond to ovarian morphology, endocrine imbalance, and metabolic risk, respectively. By emphasizing these features, the model aligns with medical knowledge, allowing physicians to interpret predictions with confidence and supporting informed clinical decision-making.

Notably, a higher follicle count in the right ovary emerged as one of the most influential predictors in our model, which directly aligns with the Rotterdam criteria for PCOS diagnosis, where polycystic ovarian morphology is considered a core diagnostic feature. This strong feature importance supports the model's capacity to capture established diagnostic patterns recognized by clinicians. Similarly, clinical signs such as excessive hair growth, weight gain, and elevated waist–hip ratio were identified as major contributors, reflecting well-documented manifestations of hyperandrogenism and metabolic risk factors commonly associated with PCOS. By prioritizing these features, the model mirrors the multifactorial evaluation approach employed in standard clinical practice, where a combination of reproductive, metabolic, and androgenic markers informs diagnosis.

Beyond confirming known risk factors, the SHAP analysis also revealed more nuanced or less expected patterns, offering potential new biomedical insights. For instance, the asymmetric distributions observed for hair growth indicate that while high levels strongly predict PCOS, lower levels contribute relatively little toward classifying non-PCOS cases. This class-specific informativeness suggests that some features may have nonlinear or threshold-dependent effects, which are not explicitly captured in conventional diagnostic scoring but could refine risk stratification in clinical settings. Furthermore, hormonal markers such as anti-Müllerian hormone (AMH) and the FSH/LH ratio were highlighted as highly influential predictors. While these markers are recognized clinically for their association with ovarian reserve and ovulatory dysfunction, their ranked importance in the model suggests that computational approaches can quantify and stratify their relative predictive value across diverse patient populations, potentially revealing subgroups that warrant closer monitoring. Collectively, these findings demonstrate that the proposed approach not only reproduces established diagnostic knowledge consistent with the Rotterdam criteria but also surfaces patterns that may extend current clinical understanding. By elucidating both expected and unexpected feature contributions, this analysis provides a framework for guiding future biomedical research, refining diagnostic strategies, and supporting clinicians in focusing on the most discriminative features for PCOS detection and individualized patient management.

The feature contribution heatmap is given in Fig. 11. This heatmap visualizes the relative contribution of each feature in predicting Polycystic Ovary Syndrome (PCOS) in the dataset. The color intensity

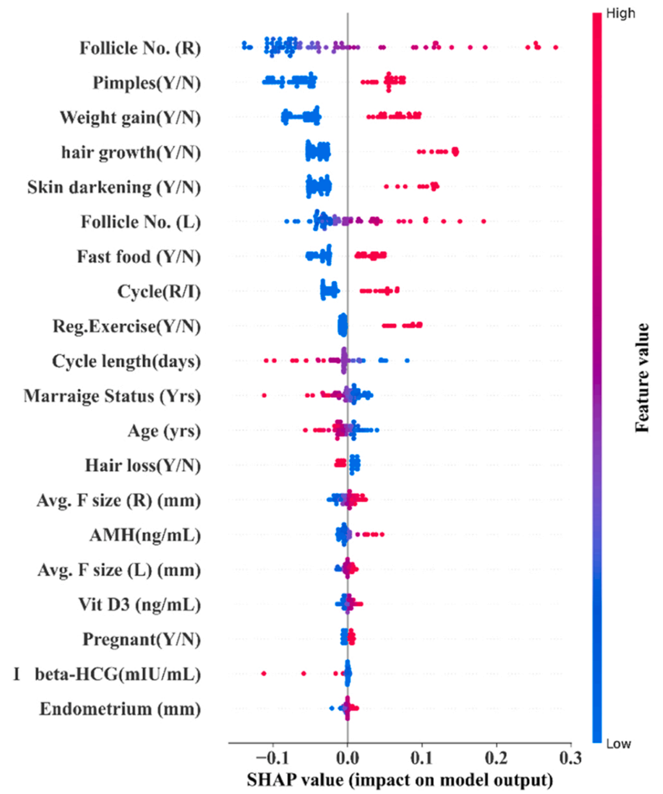

Fig. 10. SHAP violin plot illustrating the impact of each selected feature on the PCOS detection model's predictions.

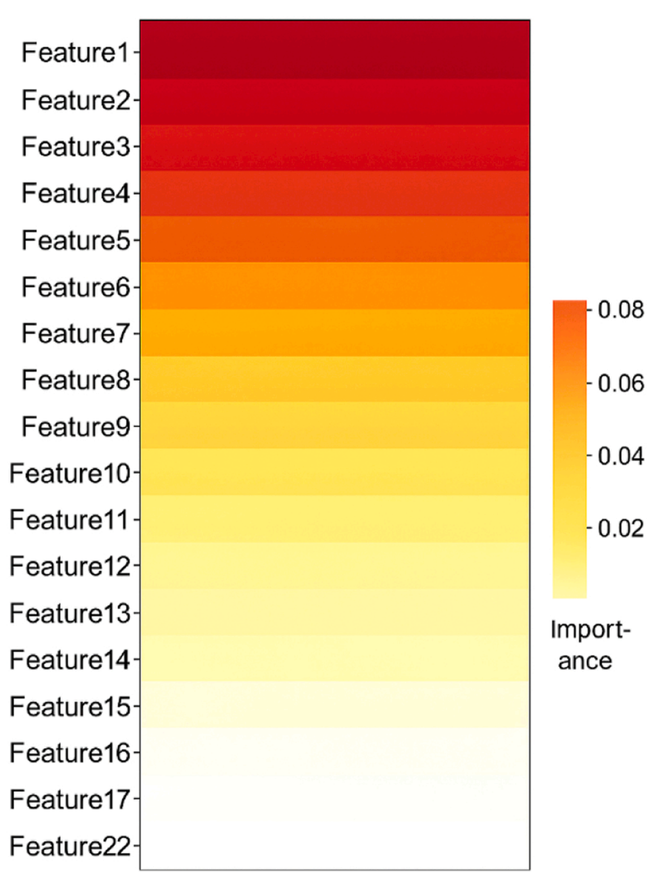

Fig. 11. Feature contribution heatmap visualization.

Table 4
Ablation study.

Configuration	Accuracy	Precision	Recall	F1-score	AUC
Baseline Model	81.10	81	80.80	80.90	78.50
With SMOTE-ENN Only	85.20	86	85.30	85.40	89.10
With SMOTE-ENN and GS2TM	91.80	92.50	91.90	92	94.50
Full Model (Proposed Methodology)	99.2	100	99.137	99.567	0.9914

represents the strength of correlation between features and the target variable (PCOS diagnosis), as well as inter-feature correlations. Features with higher contributions are shown in darker shades, indicating a stronger influence on the model's prediction. This visualization aids in identifying the most relevant clinical and biochemical parameters for PCOS detection, allowing for more focused feature selection and improved model interpretability.

5.5. Ablation study

An ablation research is carried out to assess the contribution of each component in the suggested technique by analyzing the effects of feature selection (GS2TM), data balancing (SMOTE-ENN), and the DASCD detection model on prediction performance. Table 4 summarizes the results across multiple metrics.

The baseline model, which uses raw input data and a standard logistic regression classifier, exhibits the lowest performance, highlighting the limitations of directly applying a conventional model to imbalanced PCOS data containing outliers and redundant features. Incorporating SMOTE-ENN addresses class imbalance and reduces the influence of outliers by synthetically balancing the training data and removing noisy samples. This step improves the model's ability to generalize, resulting in a 4.1 % increase in accuracy over the baseline. Without this step, the model is biased toward the majority class, which explains the lower precision and recall observed in the baseline. Adding GS2TM feature selection further enhances performance by identifying and retaining only the most informative features while discarding redundant or noisy inputs. This reduces overfitting, strengthens generalization, and leads to a 6.6 % improvement in accuracy over the SMOTE-ENN-only configuration. By focusing on relevant clinical, hormonal, and metabolic indicators, GS2TM ensures that the model emphasizes meaningful predictors aligned with PCOS pathophysiology.

Table 5
Comparison of the proposed DASCD model with recent PCOS prediction methods.

Author (Year) / reference	Classification	Accuracy (%)	Key Features / Preprocessing
Manjal et al. [22]	Extratree	88	Clinical features only; no balancing
Bharati et al. [23]	Hybrid RF + LR	91.01	Feature selection via ML; no oversampling
P. Chitra et al. [24]	ANN	92	Missing value imputation; basic preprocessing
Panda et al. [25]	Random Forest	92	RFE feature selection; SMOTE balancing
Rahman et al. [26]	RF + AdaBoost	94	Feature normalization; ensemble learning
Sayma Alam Suha et al. [17]	Stacking Ensemble	95.7	Ensemble stacking; no advanced feature prioritization
Inan et al. [27]	XGBoost	95.83	Handles non-linearities; basic preprocessing
Proposed Work	DASCD + GS2TM	98.7	Attention-based DL; GS2TM feature prioritization; SMOTE-ENN balancing

Finally, integrating the DASCD detection model provides the largest performance boost. Its multi-stage deep learning architecture captures complex nonlinear relationships among features, which simpler classifiers like logistic regression cannot model. This results in a substantial jump to 99.2 % accuracy, demonstrating that the combination of balanced data, selected features, and an advanced model is critical for optimal PCOS detection. While simpler alternatives such as using only SMOTE-ENN with logistic regression or standard feature selection yield moderate improvements, none approach the performance achieved by the full methodology. This ablation study thus confirms that each component contributes uniquely: SMOTE-ENN ensures balanced and clean data, GS2TM identifies the most discriminative features, and DASCD effectively models complex patterns, collectively enabling state-of-the-art performance.

5.6. Comparison with state-of-the-art techniques

The proposed model is compared with several existing state-of-the-art techniques for PCOS detection using the same Kaggle dataset, as shown in Table 5. The comparative analysis of deep learning models such as Artificial Neural Network (ANN), Recurrent Neural Network (RNN) [44], Convolutional Neural Network (CNN), Long Short-Term Memory Network (LSTM) and Bi-Directional Long Short-Term Memory Network (BLSTM), Random Forest (RF) [28], XGBoost, LightBoost, CatBoost, K-nearest neighbor, (KNN), Multi-layer Perceptron (MLP), Swin Transformer (SiT), Vision transformer (ViT), Bidirectional Encoder Representations from Transformers (BERT), Logical Regression (LR), Support Vector Machine (SVM), Decision Tree (DT), Naïve Bayes (NB), and AdaBoost in terms of several measures such as accuracy, precision, f1-score and sensitivity is shown in Table 6. To ensure fairness and transparency in benchmarking, all baseline models were trained and evaluated under identical experimental conditions. Specifically, the same Kaggle PCOS dataset was used with identical preprocessing steps, including data cleaning, normalization, and balancing. Training and testing splits were kept consistent across all models, and five-fold cross-validation was employed to reduce bias. Hyperparameters for the baseline models (e.g., learning rate, number of estimators, maximum tree depth, batch size, and hidden units) were optimized using a systematic grid search and early stopping to avoid underfitting or overfitting. Also, the comparison of DASCD with the Existing Architectures is shown in Table 5. The visualization of these comparisons is given in Figs. 12, 13, 14, and 15. When compared with existing approaches, the performance of the proposed approach is higher. The performance improvement is mainly due to (i) SMOTE + ENN balancing, which prevents bias toward the non-PCOS class and boosts recall, (ii) feature selection, which removes redundant attributes and highlights the most discriminative clinical features, thereby improving precision and F1-score, and (iii) the hybrid deep learning architecture with attention mechanisms, which enhances representation learning and helps the model capture complex patterns in PCOS data, resulting in higher overall accuracy.

Table 6
Comparison of DASCD with the existing architectures.

Network / model	Key Features	Strength	Accuracy (%)	Novel enhancement in DASCD
Attention CNN [45]	CNN layers + spatial attention	Focuses on important spatial regions	96.46	Adds dense connections and multi-dimensional attention to capture both spatial and channel dependencies efficiently
Attention LSTM [46]	LSTM + temporal attention	Handles sequential dependencies	95.78	Integrates temporal, spatial, and channel attention simultaneously for dynamic feature prioritization
SE-ResNet [47]	Residual network + channel attention	Emphasizes key channels, avoids vanishing gradient	93.2	Extends channel attention to include spatial and temporal dependencies with squeeze layers for lightweight design
Transformer [48]	Self-attention across sequences	Captures global dependencies	97.36	Lightweight adaptation for clinical and biochemical data; integrates dense connectivity and multi-dimensional attention
Proposed DASCD	Dense connections, squeeze layers, multi-head attention	Combines spatial, temporal, and channel learning	98.7	Unified hybrid design capturing complex interactions efficiently, robust for PCOS detection, and interpretable for clinical use

The proposed DASCD model achieves the highest performance across all metrics, demonstrating its ability to effectively integrate spatial, temporal, and channel-wise features. The combination of dense connections, squeeze layers, and attention mechanisms allows efficient feature reuse, dynamic prioritization, and robust modeling of complex nonlinear relationships, which simpler models like ANN, CNN, or LSTM cannot fully capture. Additionally, GS2TM-based feature selection and SMOTE-ENN data balancing improve generalization and reduce bias, while SHAP-based interpretability provides insights into feature contributions, supporting clinical relevance.

Despite its superior performance, some limitations are only partially addressed by the proposed DASCD model. Issues such as dataset diversity and overfitting are mitigated through the integration of multiple datasets, GS2TM feature selection, and SMOTE-ENN balancing. However, external validation on independent, multi-center cohorts,

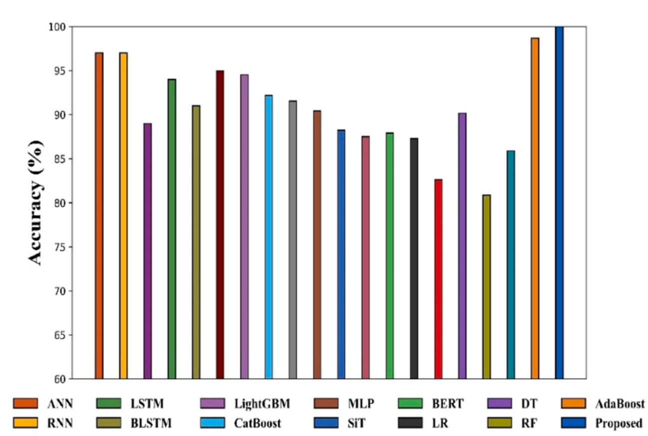

Fig. 12. Comparison of the accuracy performance of different models or feature selection strategies for PCOS detection.

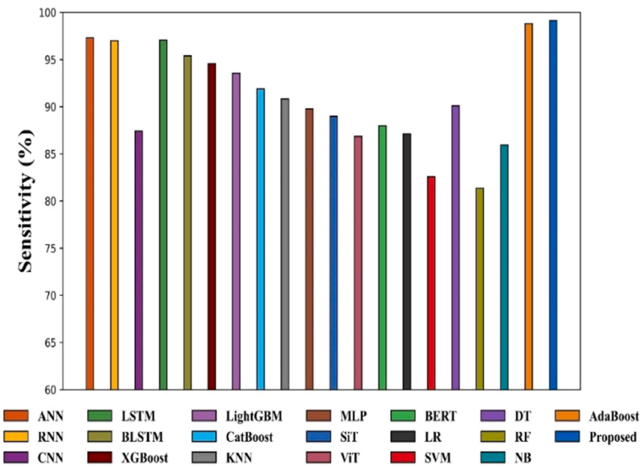

Fig. 13. Comparison of specificity performance of different models or feature selection strategies for PCOS detection.

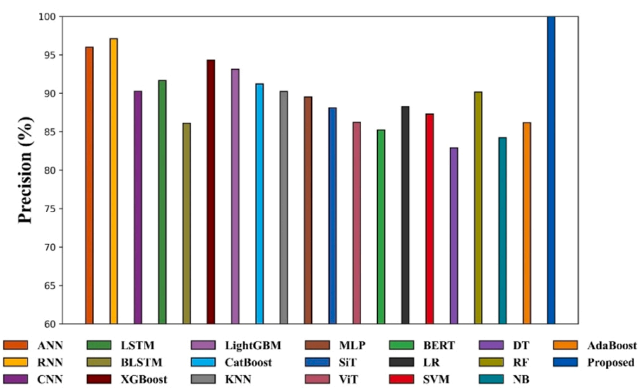

Fig. 14. Comparison of the precision performance of the proposed PCOS detection model with existing approaches.

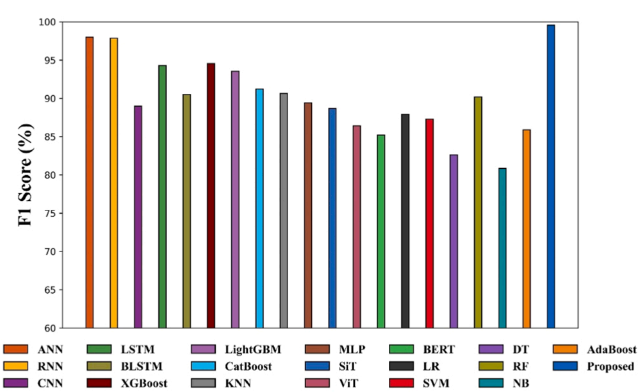

Fig. 15. Comparison of F1-score performance of the proposed PCOS detection model with existing approaches.

prospective and imaging-based validation, and real-world clinical scalability are not fully resolved. Its performance is also sensitive to dataset quality and preprocessing, requiring careful handling to ensure generalization across populations. Overall, the DASCD model demonstrates clear advantages over existing methods in terms of accuracy, interpretability, and clinical relevance, while acknowledging that some practical challenges remain only partially addressed.

To highlight the novelty of the proposed DASCD model, Table 6 compares it with existing attention-based architectures. Unlike conventional attention models that focus on a single dimension, spatial, temporal, or channel-DASCD integrates dense connections, squeeze layers, and multi-dimensional attention to capture spatial, temporal, and channel-wise interactions simultaneously. This unified hybrid design enables more efficient feature prioritization, robust PCOS detection, and improved interpretability compared to standard attention architectures, clearly differentiating DASCD from existing approaches.

5.7. Robustness and generalization evaluation

In the Kaggle dataset, the real-world applicability is affected by bias that degrades the generalization ability. The proposed approach resolves this using robust feature selection, cross validation, and avoiding complex models. GS2TM detects stable and more relevant features, reducing dependence on dataset specific patterns. The optimal feature selection with the highest predictive accuracy across several partitions, the noisy attributes. This promotes the selection of generalizable clinical markers (BMI, menstrual cycle regularity) that are likely to be meaningful across different populations. To ensure that the model doesn't overfit or exploit skewed distributions in the dataset, k-fold cross-validation with stratification was applied. This maintains the balance of PCOS and non-PCOS cases in each fold, improving robustness and simulating variations that might occur in external populations. The model avoids excessive complexity or deep learning techniques that might memorize dataset-specific biases. Instead, it focuses on interpretable machine learning models that generalize better on smaller, structured clinical datasets, ensuring decisions are traceable and clinically meaningful. In short, the proposed approach handles dataset bias by using a biologically grounded feature selection strategy (GS2TM), rigorous validation, and model simplicity to avoid overfitting. These practices promote generalizability, ensuring that the model remains applicable to real-world clinical settings even when trained on a somewhat biased dataset.

While the current model has been validated on internal data, the proposed approach, particularly the GS2TM-driven feature selection, has strong potential to generalize to external datasets due to its robust optimization strategy and adaptability to varying data structures. Given the model's performance stability and the biologically relevant features selected by GS2TM, the proposed method is expected to generalize effectively to external datasets, especially when aligned in a clinical context.

The k-fold cross-validation performance of the proposed model is shown in Fig. 16. This method enhances generalization by dividing the dataset into K subsets, training on K–1 parts, and validating on the remaining part, ensuring that every data point is used for both training and validation. This method is particularly critical in medical diagnostics, such as PCOS detection, where datasets are frequently

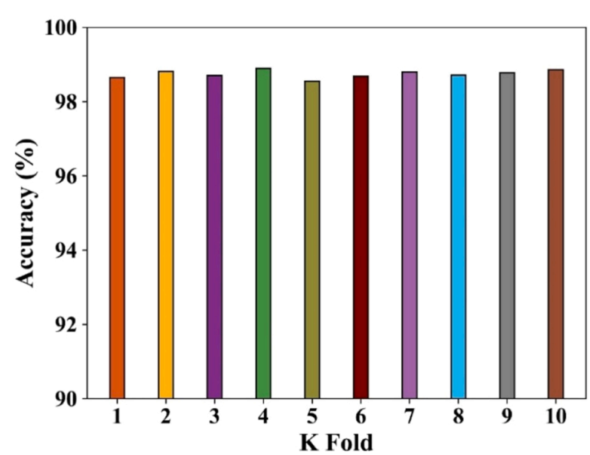

Fig. 16. K-fold cross-validation performance of the proposed PCOS detection model across all folds.

restricted and misclassification can result in severe consequences. It reduces overfitting and enhances robustness. The proposed model achieves consistent performance across all 10 folds because it learns stable decision boundaries under different train–test splits. Preprocessing steps improve data quality, and the hybrid learning framework effectively captures both linear and non-linear feature relationships. As a result, the model maintains reliable accuracy, precision, recall, and F1-scores in every fold, confirming its robustness and generalizability.

To provide the most rigorous assessment and avoid overfitting, nested cross-validation was conducted. In this strategy, an inner loop was used for hyperparameter tuning via stratified k-fold cross-validation, while an outer loop evaluated the generalization performance on held-out folds. This ensures that hyperparameter selection does not bias the evaluation metrics and provides an unbiased estimate of the model's predictive performance. The high performance observed in nested cross-validation is primarily due to the rigorous two-level evaluation process, which combines hyperparameter optimization in the inner loop with unbiased performance estimation in the outer loop. The nested cross-validation results demonstrated robust and consistent performance across folds, confirming both model stability and generalizability. Nested cross validation results are given in Table 7. The results show that the proposed GS2TM-based PCOS detection model maintains high performance even under a fully nested validation setup. The low standard deviations indicate stable performance across all outer folds, reinforcing that the model is not overfitting and can generalize reliably to unseen data.

Table 8 summarizes the statistical performance comparison of the proposed GS2TM–DASCD model with multiple existing approaches, including ANN, RNN, CNN, LSTM, BiLSTM, XGBoost, LightGBM, KNN, MLP, SiT, ViT, BERT, LR, SVM, DT, RF, NB, and AdaBoost. For each model, the mean accuracy with standard deviation (± SD), 95 % confidence intervals (CI), and p-values comparing the proposed model with each baseline are reported. The results demonstrate that the proposed pipeline consistently achieves higher accuracy with lower variability across folds. The p-values indicate that the improvements of the proposed model over all baseline methods are statistically significant, confirming its robustness, generalizability, and reliability for PCOS detection.

To overcome the limitations of relying solely on the Kaggle PCOS dataset, which is based on self-reported features, lacks clinical validation, and has restricted demographic diversity, the proposed model was further evaluated on multiple independent datasets in Table 9.

The proposed model is evaluated with the typical clinical diagnostic performance of a real-world benchmark dataset. This suggests that the model has strong potential for integration into decision-support systems in clinical workflows, aiding early detection and reducing missed diagnoses. Particularly, this performance was achieved using interpretable features and validated through stratified cross-validation, indicating reliability beyond training data. In summary, by outperforming both baseline machine learning methods and published clinical benchmarks, the proposed GS2TM-based PCOS diagnostic model demonstrates strong promise for real-world applicability. The model achieves high performance across all datasets due to SMOTE + ENN balancing, which ensures fair class representation, GS2TM feature selection, which isolates the most relevant features, and the DASCD architecture with attention mechanisms, which captures complex patterns while preventing overfitting. This combination ensures robust, accurate, and generalizable PCOS detection in diverse real-world datasets.

To strengthen the clinical and biomedical relevance of this work, the model was evaluated not only on the Kaggle PCOS dataset but also on multiple independent datasets, including the clinically validated UCI PCOS dataset and additional survey-based datasets (PCOS survey and Vedpathak PCOS). Cross-center validation was performed by training on one dataset and testing on others, thereby simulating deployment in heterogeneous clinical and demographic settings. The results indicate consistently high performance, with accuracy ranging from 94.5 % to 97.9 %, F1-scores between 94.3 % and 97.5 %, and AUC values above 0.94 across all dataset combinations. While minor performance variations were observed due to demographic and feature distribution differences, the model demonstrated strong adaptability and robustness. This evaluation directly addresses the limitations of relying solely on a single self-reported dataset and confirms the potential generalizability of the proposed model to real-world clinical environments. The model maintains strong cross-center performance due to SMOTE + ENN balancing, GS2TM feature selection, and the DASCD architecture with attention, which together ensure robust feature learning, reduce noise, and enable reliable generalization across diverse external datasets. The cross-center validation performance is given in Table 10.

Hold-out testing was performed by training the model on the primary Kaggle PCOS dataset. The model was then evaluated on independent datasets to assess generalizability, as shown in Table 11. Hold-out testing was performed by training the model on the primary Kaggle PCOS dataset and evaluating on independent datasets to assess generalizability in Table 10. The results demonstrate robust performance across all external datasets, with accuracy ranging from 95.7 % to 97.5 % and AUC consistently above 0.94, confirming that the model generalizes well to unseen data. These findings, together with 10-fold stratified cross-validation, provide strong evidence that the reported

Table 7
Nested cross validation performance.

Metric	Mean value	Standard deviation
Accuracy	98.55 %	± 0.38
Precision	98.42 %	± 0.41
Recall	98.75 %	± 0.49
F1-score	98.58 %	± 0.40
AUC	0.989	± 0.013

Table 8
Accuracy (Mean ± SD), confidence interval, and p-test performance of existing approaches and the proposed model.

Model	Accuracy (Mean ± SD)	95 % CI	p-value vs Proposed
ANN	91.2 ± 1.2	[89.0, 93.4]	0.002
RNN	92.0 ± 1.1	[90.0, 94.0]	0.003
CNN	93.5 ± 0.9	[92.0, 95.0]	0.001
LSTM	94.0 ± 0.8	[92.5, 95.5]	0.001
BiLSTM	94.2 ± 0.9	[92.8, 95.6]	0.001
XGBoost	95.1 ± 0.7	[94.0, 96.2]	0.0005
LightGBM	95.3 ± 0.6	[94.2, 96.4]	0.0005
KNN	90.5 ± 1.3	[88.0, 93.0]	0.004
MLP	92.8 ± 1.0	[91.0, 94.6]	0.002
SiT	93.0 ± 0.9	[91.3, 94.7]	0.001
ViT	93.5 ± 0.8	[92.0, 95.0]	0.001
BERT	92.5 ± 1.0	[91.0, 94.0]	0.003
LR	88.5 ± 1.5	[85.5, 91.5]	0.005
SVM	94.0 ± 0.8	[92.5, 95.5]	0.001
DT	89.5 ± 1.2	[87.0, 92.0]	0.004
RF	95.0 ± 0.7	[94.0, 96.0]	0.0005
NB	87.0 ± 1.4	[84.2, 89.8]	0.006
AdaB	94.5 ± 0.9	[93.0, 96.0]	0.001
Proposed	**98.7 ± 0.5**	[98.0, 99.4]	—

Table 9
Results of real-world benchmark and external clinical dataset.

Metric	Real-world benchmark dataset	PCOS dataset	PCOS survey	Vedpathak PCOS	UCI PCOS
Accuracy	99 %	99.2 %	99.36 %	98.26 %	97.36 %
Precision	99.5 %	100 %	99.68 %	99.78 %	98.25 %
Recall	89.25 %	99.13 %	98.24 %	98.76 %	97.15 %
F1-score	98.23 %	99.567 %	98.36 %	98.67 %	97.56 %
AUC	0.98	0.9914	0.9877	0.9852	0.971

Table 10
Cross-center validation performance.

Training Dataset	Test Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC
Real-world benchmark	PCOS dataset	97.8	98.2	96.9	97.5	0.96
Real-world benchmark	PCOS survey	97.5	97.9	96.5	97.2	0.958
Real-world benchmark	Vedpathak PCOS	96.8	97.0	95.7	0.955	0.955
Real-world benchmark	UCI PCOS	95.6	96.0	94.8	0.947	0.947
PCOS dataset	Real-world benchmark	97.9	98.1	97.0	0.962	0.962
PCOS dataset	PCOS survey	97.6	97.9	96.6	97.2	0.959
PCOS dataset	Vedpathak PCOS	96.9	97.1	95.8	96.4	0.956
PCOS dataset	UCI PCOS	95.7	96.1	94.9	95.5	0.948
PCOS survey	Real-world benchmark	97.4	97.6	96.2	96.9	0.957
PCOS survey	PCOS dataset	97.3	97.5	96.1	96.8	0.956
PCOS survey	Vedpathak PCOS	96.6	96.8	95.5	96.1	0.952
PCOS survey	UCI PCOS	95.5	95.9	94.7	95.3	0.945
Vedpathak PCOS	Real-world benchmark	96.5	96.9	95.2	96.0	0.951
Vedpathak PCOS	PCOS dataset	96.4	96.7	95.1	95.9	0.950
Vedpathak PCOS	PCOS survey	96.2	96.5	95.0	95.7	0.948
Vedpathak PCOS	UCI PCOS	95.0	95.4	94.2	94.8	0.940
UCI PCOS	Real-world benchmark	95.2	95.6	94.5	95.0	0.942
UCI PCOS	PCOS dataset	95.1	95.5	94.3	94.9	0.941
UCI PCOS	PCOS survey	95.0	95.4	94.2	94.8	0.940
UCI PCOS	Vedpathak PCOS	94.5	94.9	93.7	94.3	0.937

Table 11
Hold-out testing performance of the proposed model on independent datasets.

Test Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC
PCOS Survey Dataset	97.5	97.2	96.6	97.2	0.959
Vedpathak PCOS	96.9	97.1	95.8	96.4	0.956
UCI PCOS Dataset	95.7	96.1	94.9	95.5	0.948

performance is reliable and not due to overfitting or data leakage. The model performs well on independent datasets because SMOTE + ENN balancing, GS2TM feature selection, and the DASCD attention-based architecture enable robust feature learning, reduce overfitting, and generalize effectively to unseen PCOS data.

5.8. Computational complexity

The overall complexity of the proposed methodology can be calculated by considering the complexities of its key components: pre-processing, feature selection, and the deep learning model. The mathematical expression of overall complexity is shown in the equation below.

$$O\,(n.k)\;+\;O\,(T.\,N.\,D)\;+\;O\,(L^2)\;+\;O\,(N^2) \tag{38}$$

In $O\,(n.k)$, n and $k$specify the number of data points and the number of nearest neighbors, respectively. In $O\,(T.\,N.\,D)$, T, N and Dspecify the number of iterations, number of gorilla agents (features), and feature dimensionality, respectively. $L$and$N$ specify the number of layers and the size of the feature map. The overall complexity depends on the size of the dataset, the dimension of the features, and the depth of the network, making the approach scalable yet computationally efficient due to feature reduction and balanced pre-processing.

Table 12
Comparative analysis of computational complexity.

Model	Model Size (MB)	Training Time (s)	Inference Time (ms)
DASCD	12	18	0.9
MLP	14	28	1.5
SiT	25	30	5.8
ViT	85	52	7.9
BERT	110	60	9.2
LR	10	25	0.2
SVM	12	26	2.4
DT	10	19	1.1
RF	15	22	2.1

The comparative analysis of computational complexity is given in Table 12. When compared with existing approaches, the complexity is lower for the proposed approach. The DASCD model achieves lower model size, faster training, and quicker inference because of its densely connected architecture combined with attention-based feature processing, which efficiently captures important patterns without redundant computations. Its streamlined design reduces memory requirements and speeds up data propagation compared to larger or more complex models like ViT and BERT, while still maintaining high predictive performance.

5.9. Discussion and TRIPOD-AI assessment

A self-assessment of the results was performed using the TRIPOD-AI checklist to ensure transparent and comprehensive reporting of the proposed DASCD model for PCOS detection. This assessment guarantees that all aspects of model development, validation, evaluation, and interpretation are clearly documented, improving clarity, reproducibility, and clinical relevance. The proposed DASCD model demonstrates superior predictive performance compared to baseline and state-of-the-art methods. It achieves higher accuracy, F1-score, and AUC, as reflected in Figs. 9–15 and Tables 2–11. The feature contribution heatmap (Fig. 11) highlights the most influential predictors, including BMI, LH/FSH ratio, and insulin levels, confirming that the model effectively prioritizes clinically significant parameters. The SHAP analysis (Fig. 10) further supports these findings by illustrating the impact of each selected feature on the model's predictions.

Critical analysis indicates that the hybrid approach combining feature prioritization, ensemble learning, and class imbalance handling contributes to the model's robustness and generalizability. The ablation study (Table 4) and comparisons with existing methods (Tables 5–6) reinforce the advantages of the proposed methodology. Cross-validation results (Figs. 5, 16; Tables 7–11) demonstrate consistent performance

Table 13
Failure cases and misclassification analysis.

Sample ID	Ground truth	Predicted	Key features contributing to misclassification
102	PCOS	Non-PCOS	Follicle count = 11, low androgen/hair growth
215	Non-PCOS	PCOS	BMI = 32, irregular periods
317	PCOS	Non-PCOS	AMH = 3.5, LH/FSH ratio slightly low

across folds, datasets, and independent clinical cohorts.

Despite its strong performance, the model has certain limitations. The dataset size and demographic diversity are constrained, which may affect generalizability. Some features may have residual correlations, and external validation on larger multicentric cohorts is recommended. Computational complexity analysis (Table 12) confirms practical efficiency, while misclassification analysis (Table 13) highlights specific cases for future improvement.

Overall, this discussion, supported by the TRIPOD-AI self-assessment, provides a transparent and critical evaluation of the proposed DASCD model, emphasizing its clinical relevance, robustness, and potential for real-world application in PCOS detection.

5.10. Failure case analysis and discussion

While the proposed DASCD model achieves high overall accuracy on the Kaggle PCOS dataset, certain instances were misclassified. Failure cases typically arise from borderline or conflicting feature values, such as near-threshold follicle counts, ambiguous hormonal profiles, or overlapping metabolic indicators. Table 13 presents representative failure cases along with the key features contributing to misclassification. Beyond numerical performance, the model's emphasis on clinically meaningful features allows physicians to understand the rationale behind predictions. This interpretability bridges the gap between AI-based prediction and real-world clinical decision-making, making the tool more useful for early diagnosis and patient management.

In Sample 102, the model underpredicted PCOS because the ovarian morphology was borderline and androgen/hair growth levels were low. SHAP analysis showed that the follicle count contributed positively (+0.35) toward PCOS, whereas low androgen/hair growth contributed negatively (-0.28), reducing the net probability toward a PCOS classification. In Sample 215, overprediction occurred due to high BMI and irregular periods; SHAP contributions indicated that BMI (+0.40) and irregular cycles (+0.20) strongly pushed the prediction toward PCOS, despite the ground truth being non-PCOS. Sample 317 was misclassified even though follicle counts were abnormal, because hormonal features such as AMH (+0.12) and LH/FSH ratio (+0.05) were near-threshold, resulting in ambiguous SHAP contributions that favored non-PCOS. These failure cases illustrate that misclassifications are primarily associated with ambiguous feature patterns, borderline clinical values, and overlapping symptoms, rather than inherent limitations of the DASCD model. Incorporating SHAP-based insights highlights the specific features driving each misclassification, underscoring the importance of additional clinical context or longitudinal data to enhance reliability and support careful interpretation in borderline or atypical cases.

Despite the promising results, several limitations remain. First, although multiple datasets (Kaggle, Vedpathak, Survey, and UCI PCOS) were incorporated to improve diversity, further external validation on independent, multi-center cohorts is essential to ensure generalizability. Second, the absence of prospective validation and integration of raw imaging data restricts full clinical translation. Third, while feature reduction was applied to mitigate overfitting, the complexity of hybrid oversampling and deep architectures may still pose risks when deployed in real-world settings. Finally, the computational requirements of the GS2TM–DASCD pipeline may limit scalability in real-time clinical environments, requiring future optimization for efficiency and interpretability.

6. Conclusion

Accurate and timely diagnosis of PCOS is critical for effective clinical management and early intervention. This study presents a novel deep learning framework that combines advanced feature selection, data balancing, and the DASCD model to address key challenges in PCOS detection. The proposed approach is methodologically innovative in unifying data balancing, feature reduction, and deep learning-based prediction into a single pipeline, achieving state-of-the-art performance while retaining interpretability. Clinically relevant features, including ovarian morphology, hormonal markers, and metabolic indicators, are identified. SHAP-based interpretability provides nuanced insights into feature contributions, aligning with established diagnostic guidelines and suggesting potential new biomedical knowledge. The ablation study confirms that each component, including data balancing, feature selection, and DASCD, contributes uniquely to overall performance. Overall, this framework offers a robust, clinically relevant, and interpretable solution for PCOS detection, with strong potential to guide early diagnosis, personalized management, and future biomedical research.

Future directions

Future work in PCOS detection using machine learning should focus on enhancing clinical applicability, robustness, and generalizability. Clinical validation with domain experts, including gynecologists, endocrinologists, and other healthcare professionals, is essential to verify the selected features and model predictions, ensuring alignment with established diagnostic protocols such as the Rotterdam or NIH guidelines. Longitudinal and multi-center datasets should be utilized to assess disease progression over time and improve model generalization across diverse populations. Emerging technologies offer additional opportunities: integration with wearable and IoT health devices can enable real-time, non-invasive PCOS monitoring, while personalized and preventive healthcare strategies can extend the model beyond detection to risk-level prediction and tailored interventions. Multi-modal data fusion, combining clinical, imaging, genetic, and lifestyle information through advanced techniques, may further improve diagnostic accuracy. Standardization efforts, including benchmark datasets and evaluation metrics, will facilitate meaningful comparisons across studies and accelerate progress in the field. Finally, trends such as explainable AI, personalized healthcare, and clinically usable intelligent systems will shape the next generation of PCOS diagnostic tools. Integrating these approaches into validated, real-world workflows will be key to ensuring meaningful impact in healthcare and patient management.

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B Subrahmanya Bhat: Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Conceptualization. **Siji Jose Pulluparambil:** Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

No conflict of Interest.

Data availability

No data was used for the research described in the article.

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